



4. Comprehensive Medical Evaluation and Assessment of Comorbidities: Standards of Care in Diabetes—2026

Diabetes Care 2026;49(Suppl. 1):S61–S88 | <https://doi.org/10.2337/dc26-S004>

American Diabetes Association
Professional Practice Committee for
Diabetes*

The American Diabetes Association (ADA) “Standards of Care in Diabetes” includes the ADA’s current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA’s clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

PERSON-CENTERED COLLABORATIVE CARE

Recommendations

4.1 A communication style that uses person-centered, culturally sensitive, and strength-based language and active listening; elicits individual preferences and beliefs; and assesses literacy, numeracy, and potential barriers to care should be used to optimize health outcomes and health-related quality of life. **B**

4.2 People with diabetes can benefit from a coordinated interprofessional team that may include but is not limited to diabetes care and education specialists, primary care and subspecialty clinicians, nurses, registered dietitian nutritionists, exercise specialists, pharmacists, dentists, podiatrists, and behavioral health professionals. **C**

A successful medical evaluation and care plan depends on the beneficial interactions and care coordination between the person with diabetes and the care team. The Chronic Care Model (1) and other evidence-based care delivery models (see section 1, “Improving Care and Promoting Health in Populations”) can support a person-centered approach to care that requires a close working relationship between the person with diabetes, their clinicians, and the broader health care team involved in treatment planning and implementation. People with diabetes should receive health care from a coordinated interprofessional team that may include but is not limited to diabetes care and education specialists, primary care and subspecialty clinicians, nurses, registered dietitian nutritionists, exercise specialists, pharmacists, dentists, podiatrists, behavioral health professionals, and community

*A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at <https://doi.org/10.2337/dc26-SINT>.

Duality of interest information for each contributor is available at <https://doi.org/10.2337/dc26-SDIS>.

Suggested citation: American Diabetes Association Professional Practice Committee for Diabetes. 4. Comprehensive medical evaluation and assessment of comorbidities: Standards of Care in Diabetes—2026. *Diabetes Care* 2026;49(Suppl. 1):S61–S88

The BONE HEALTH subsection has received endorsement from the American Society for Bone and Mineral Research.

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partners such as community health workers and community paramedics. Individuals with diabetes and their care partners or caregivers must assume an active role in their care. Based on the preferences and values of the person with diabetes, elicited by the care team, the person with diabetes, their family or support group, and the health care team together formulate the management plan, which includes lifestyle management (see section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes”) and pharmacotherapy, as appropriate.

The goals of treatment for diabetes are to prevent or delay complications and optimize quality of life (Fig. 4.1). Treatment goals and plans should be cocreated by the care team and people with diabetes based on their individual

preferences, values, and goals. This individualized management plan should take into account the person's age, cognitive abilities, school/work schedule and conditions, health beliefs, support systems, eating patterns, physical activity, social situation, financial concerns, cultural factors, literacy and numeracy (mathematical literacy), diabetes history (duration, complications, and current use of medications), comorbidities, disabilities, health priorities, other medical conditions, preferences for care, access to health care services, and life expectancy. People living with diabetes should be engaged in conversation about these aspects of their lives and diabetes management, with routine reassessment as necessary given their changing circumstances across the life span. Various strategies and techniques

should be used to support the person's self-management efforts, including providing education on problem-solving and coping skills for all aspects of diabetes management.

Communication between health care professionals and people with diabetes and their families should acknowledge that multiple factors impact diabetes management but also emphasize that collaboratively developed treatment plans and a healthy lifestyle can significantly improve disease outcomes and well-being (2). Thus, the goal of communication between health care professionals and people with diabetes is to establish a collaborative relationship and to assess and address self-management barriers without blaming people with diabetes for “noncompliance” or “nonadherence”

Decision cycle for person-centered glycemic management in type 2 diabetes



Figure 4.1—Decision cycle for person-centered glycemic management in type 2 diabetes. BGM, blood glucose monitoring; BP, blood pressure; CGM, continuous glucose monitoring; CKD, chronic kidney disease; CVD, cardiovascular disease; DSMES, diabetes self-management education and support; HF, heart failure. MASLD, metabolic dysfunction-associated steatotic liver disease. Adapted from Davies et al. (307).

when the outcomes of self-management are not optimal (3). The familiar terms noncompliance and nonadherence denote a passive, obedient role for a person with diabetes in “following doctor’s orders,” which is at odds with the active role people with diabetes take in the day-to-day decision-making, planning, monitoring, evaluation, and problem-solving involved in diabetes self-management. Using a nonjudgmental approach that normalizes periodic lapses in management may help minimize the person’s resistance to reporting problems with self-management. Empathizing and using active listening techniques, such as open-ended questions, reflective statements, and summarizing what the person said, can help facilitate communication. Perceptions of people with diabetes about their own ability, or self-efficacy, to self-manage diabetes constitute one important psychosocial factor related to improved diabetes self-management and treatment outcomes in diabetes (4) and should be a goal of ongoing assessment, education, and treatment planning.

Language has a strong impact on perceptions and behavior. Empowering language can help to inform and motivate, while shame and judgement can be discouraging. The American Diabetes Association (ADA) and the Association of Diabetes Care & Education Specialists (ADCES) (formerly called the American Association of Diabetes Educators) joint consensus report, “The Use of Language in Diabetes Care and Education,” provides the authors’ expert opinion regarding the use of language by health care professionals when speaking or writing about diabetes for people with diabetes or for professional audiences (3,5). Although further research is needed to address the impact of language on diabetes outcomes, the report includes five key consensus recommendations for language use:

- Use language that is neutral, non-judgmental, and based on facts, actions, physiology, or biology.
- Use language free from stigma.
- Use language that is strength based, respectful, and inclusive and that imparts hope.
- Use language that fosters collaboration between people with diabetes and health care professionals.

- Use language that is person centered (e.g., “person with diabetes” is preferred over “diabetic”).

COMPREHENSIVE MEDICAL EVALUATION

Recommendations

4.3 A complete medical evaluation should be performed at the initial visit and follow-up, as appropriate, to:

- Confirm the diagnosis and classify diabetes. **A**
- Assess glycemic status and previous and current treatment. **A**
- Evaluate for diabetes complications, potential comorbid conditions, and overall health status. **A**
- Identify care partners, support systems, and available resources. **E**
- Assess social determinants of health and structural barriers to optimal health and health care. **A**
- Review risk factor management in the person with diabetes. **A**
- Begin engagement with the person with diabetes in the formulation of a care management plan including goals of care. **A**
- Develop a plan for continuing care. **A**

4.4 Ongoing management should be guided by the assessment of overall health and functional status, diabetes complications, cardiovascular risk, hypoglycemia risk, and shared decision making to set therapeutic goals. **B**

The comprehensive medical evaluation includes the initial and follow-up evaluations, which comprise assessment of complications, psychosocial assessment, management of comorbid conditions, overall health, functional and cognitive status, and engagement of the person with diabetes throughout the process. While a comprehensive list is provided in **Table 4.1**, in clinical practice the health care professional may need to prioritize select components of the medical evaluation given the available resources and time. Engaging other members of the health care team can also support comprehensive diabetes care. The goal of these recommendations is to provide the health care team with guidance and information so it can optimally support people with diabetes and their care partners. In addition to the medical history, physical examination, and

laboratory tests, health care professionals should assess diabetes self-management behaviors, nutrition, social determinants of health, and psychosocial health (see section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes”) and give guidance on routine immunizations. The assessment of sleep pattern and duration, bone health, and sexual health should also be considered as an important part of the comprehensive evaluation of a person with diabetes. Follow-up visits should occur at least every 3–6 months individualized to the person and then at least annually.

Lifestyle management and behavioral health care are cornerstones of diabetes management. People with diabetes should be referred for diabetes self-management education and support, medical nutrition therapy, and assessment of behavioral health concerns as appropriate. People with diabetes should receive recommended preventive care services (e.g., immunizations and age- and sex-appropriate cancer screening); tobacco use cessation counseling; and ophthalmological, dental, podiatric, and other referrals, as needed.

The assessment of risk of acute and chronic diabetes complications and treatment planning are key components of initial and follow-up visits (**Table 4.2**). The risk of atherosclerotic cardiovascular disease (CVD) and heart failure (see section 10, “Cardiovascular Disease and Risk Management”), chronic kidney disease (CKD) staging (see section 11, “Chronic Kidney Disease and Risk Management”), presence of retinopathy and neuropathy (see section 12, “Retinopathy, Neuropathy, and Foot Care”), and risk of treatment-associated hypoglycemia should be used to individualize goals for glycemia (see section 6, “Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises”), blood pressure, and lipids and to select specific glucose-lowering medication(s) (see section 9, “Pharmacologic Approaches to Glycemic Treatment”), antihypertension medications, and lipid lowering treatment intensity.

Additional referrals should be arranged as necessary (**Table 4.2**). Clinicians should ensure that people with diabetes are appropriately screened for complications, comorbidities, and treatment burden. Glycemic management is a part, not the sole goal, of the clinical encounter.

Table 4.1—Components of the comprehensive diabetes medical evaluation at initial, follow-up, and annual visits

	Visit		
	Initial	Every follow-up	Annual
Past medical and family history			
Diabetes history			
• Characteristics at onset (e.g., age and symptoms and/or signs)	✓		
• Review of previous treatment plans and response	✓		
• Assess frequency, cause, and severity of past hospitalizations	✓		
Family history			
• Family history of diabetes in a first-degree relative	✓		
• Family history of autoimmune disorders	✓		
Personal history of complications and common comorbidities			
• Common comorbidities (e.g., obesity, OSA, and MASLD)	✓		✓
• High blood pressure or abnormal lipids	✓		✓
• Macrovascular and microvascular complications	✓		✓
• Hypoglycemia: awareness, frequency, causes, and timing of episodes	✓	✓	✓
• Presence of hemoglobinopathies or anemias	✓		✓
• Last dental visit	✓		✓
• Last foot exam	✓		✓
• Last dilated eye exam	✓		✓
• Visits to specialists	✓		✓
• Disability assessment and use of assistive devices (e.g., physical, cognitive, vision and auditory, history of fractures, and podiatry)	✓	✓	✓
• Personal history of autoimmune disease	✓		
Surgical and procedure history			
• Surgeries (e.g., metabolic surgery and transplantation)	✓	✓	✓
Interval history			
• Changes in medical or family history since last visit		✓	✓
Behavioral factors			
• Physical activity, sleep behaviors, eating patterns and weight history	✓	✓	✓
• Assess familiarity with carbohydrate counting (e.g., type 1 diabetes or type 2 diabetes treated with MDI)	✓		✓
• Screen for OSA	✓	✓	✓
• Tobacco, alcohol, and substance use	✓		✓
Medications and vaccinations			
• Current medication plan	✓	✓	✓
• Medication-taking behavior, including rationing of medications and/or medical equipment	✓	✓	✓
• Medication intolerance or side effects	✓	✓	✓
• Complementary and alternative medicine use	✓	✓	✓
• Vaccination history and needs	✓		✓
Technology use			
• Assess use of health apps, online education, patient portals, etc.	✓	✓	✓
• Glucose monitoring (meter/CGM): results and data use	✓	✓	✓
• Review insulin pump settings and use and connected pen and glucose data	✓	✓	✓
Social life assessment			
Social network			
• Identify existing social supports	✓		✓
• Identify surrogate decision maker and advanced care plan	✓		✓
• Identify social determinants of health (e.g., food security, housing stability, transportation access, financial security, and community safety)	✓		✓

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IMMUNIZATIONS**Recommendation**

4.5 Provide routinely recommended vaccinations for children, adolescents, and adults with diabetes as indicated by age (see **Table 4.3**). **A**

Children, adolescents, and adults with diabetes should receive vaccinations according to age-appropriate recommendations (6,7). The Centers for Disease Control and Prevention (CDC) provides vaccination schedules specifically for children, adolescents, and adults with diabetes ([cdc.gov/vaccines/](https://www.cdc.gov/vaccines/)). The CDC Advisory Committee on Immunization Practices (ACIP) makes recommendations based on its own review and rating of the evidence, provided in **Table 4.3** for selected vaccinations. The ACIP evidence review has evolved over time with the adoption of Grading of Recommendations Assessment, Development, and Evaluation (GRADE) in 2010 and then the Evidence to Decision or Evidence to Recommendation frameworks in 2020 (8). Here, we discuss the particular importance of specific vaccines.

COVID-19

People with underlying medical conditions, including diabetes, are more likely to become severely ill with coronavirus disease 2019 (COVID-19). COVID-19 vaccination using an appropriate number of doses of updated vaccines is recommended for everyone aged 6 months and older in the U.S. (6,7).

Hepatitis B

Compared with the general population, people with type 1 or type 2 diabetes have higher rates of hepatitis. Because of the higher likelihood of transmission of the disease, hepatitis B vaccine is recommended for adults with diabetes aged <60 years. For adults aged ≥60 years, hepatitis B vaccine may be administered at the discretion of the treating clinician based on the person's likelihood of acquiring hepatitis B infection (6,7).

Influenza

Influenza is a common, preventable infectious disease associated with high mortality and morbidity in vulnerable populations, including youth, older adults, and people with chronic diseases. Influenza vaccination

Table 4.1—Continued

	Visit		
	Initial	Every follow-up	Annual
Assess daily routine and environment, including school or work schedules and ability to engage in diabetes self-management	✓	✓	✓
Physical examination			
Height, weight, and BMI; growth and pubertal development in children and adolescents	✓	✓	✓
Blood pressure determination	✓	✓	✓
Orthostatic blood pressure measures (when indicated)	✓		✓
Fundoscopy examination (refer to eye specialist)	✓		✓
Thyroid palpation	✓		✓
Skin examination (e.g., acanthosis nigricans, insulin injection or insertion sites, and lipodystrophy)	✓	✓	✓
Comprehensive foot examination, determination of temperature, vibration or pinprick sensation, and 10-g monofilament exam	✓		✓
Visual inspection (e.g., skin integrity, callous formation, foot deformity or ulcer, and toenails)*	✓	✓	✓
Check pedal pulses and screen for PAD with ABI testing if a PAD diagnosis would change management	✓		✓
Screen for depression, anxiety, diabetes distress, fear of hypoglycemia, and disordered eating	✓		✓
Assessment for cognitive performance if indicated	✓		✓
Assessment for functional performance if indicated	✓		✓
Assessment for bone health (e.g., loss of height and kyphosis)	✓		✓
Laboratory evaluation			
A1C, if the results are not available within the past 3 months or if earlier assessment is necessary	✓	✓	✓
Lipid profile, including total, LDL, and HDL cholesterol and triglycerides‡	✓		
Liver function tests (i.e., FIB-4)‡	✓		✓
Spot urinary albumin-to-creatinine ratio	✓		✓
Serum creatinine and estimated glomerular filtration rate§	✓		✓
Thyroid-stimulating hormone in people with type 1 diabetes‡	✓		✓
Celiac disease screening in people with type 1 diabetes	✓		
Vitamin B12 if taking metformin for >5 years	✓		✓
CBC with platelets	✓		✓
Serum potassium levels in people treated with ACE inhibitors, ARBs, or diuretics§	✓		✓
Calcium, vitamin D, and phosphorous as appropriate	✓		✓

ABI, ankle brachial index; ARBs, angiotensin receptor blockers; CBC, complete blood count; CGM, continuous glucose monitor; FIB-4, fibrosis-4 index; MASLD, metabolic-associated steatotic liver disease; MDI, multiple daily injections; OSA, obstructive sleep apnea; PAD, peripheral arterial disease. *Should be performed at every visit in people with diabetes with sensory loss, previous foot ulcers, or amputations. †At 65 years of age or older. ‡May also need to be checked after initiation or dose changes of medications that affect these laboratory values (i.e., diabetes medications, blood pressure medications, cholesterol medications, or thyroid medications). §May be needed more frequently in people with diabetes with known chronic kidney disease or with changes in medications that affect kidney function and serum potassium (see Table 11.2). ||In people with presence of gastrointestinal symptoms, signs, laboratory manifestations, or clinical suspicion suggestive of celiac disease.

cardiovascular events (10). Given the benefits of the annual influenza vaccination, it is recommended for all individuals ≥ 6 months of age who do not have a contraindication. The live attenuated influenza vaccine, which is delivered by nasal spray, is an option for people who are 2–49 years of age and are not pregnant, but people with chronic conditions such as diabetes are cautioned against taking the live attenuated influenza vaccine and are instead recommended to receive the inactive or recombinant influenza vaccination. For the 2025–2026 season, all influenza vaccines offered in the U.S. are trivalent (6,7).

Pneumococcal Pneumonia

Like influenza, pneumococcal pneumonia is a common, preventable disease. People with diabetes are at increased risk for pneumococcal infection and have been reported to have a high risk of hospitalization and death, with a mortality rate as high as 50% (11). All people with diabetes should receive one of the CDC-recommended pneumococcal vaccines (12).

Respiratory Syncytial Virus

Respiratory syncytial virus (RSV) is a cause of respiratory illness in some individuals, including older adults. People with chronic conditions such as diabetes have a higher risk of severe illness. The U.S. Food and Drug Administration (FDA) approved the first vaccines for prevention of RSV-associated lower respiratory tract disease in adults aged ≥ 60 years. The ACIP recommends that all adults aged ≥ 75 years and adults aged 60–74 years who are at increased risk for severe RSV should receive a single dose of RSV vaccine (13).

ASSESSMENT OF COMORBIDITIES

Besides assessing diabetes-related complications, clinicians and people with diabetes need to be aware of common comorbidities that affect people with diabetes and that may complicate its management (14–16). Diabetes comorbidities are conditions that affect people with diabetes more often than age-matched people without diabetes. This section discusses the common comorbidities observed in people with diabetes but is not necessarily inclusive of all the conditions that have been reported.

in people with diabetes has been found to significantly reduce influenza and diabetes-related hospital admissions (9). In people

with diabetes, the influenza vaccine has been associated with lower risk of all-cause mortality, cardiovascular mortality, and

Table 4.2—Essential components for assessment, planning, and referral**Assessing risk of diabetes complications**

- ASCVD and heart failure history
- ASCVD risk factors and 10-year ASCVD risk assessment
- Staging of chronic kidney disease (see **Table 11.2**)
- Hypoglycemia risk (see section 6, “Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises”)
- Assessment for retinopathy
- Assessment for neuropathy
- Assessment for MASLD and MASH

Goal setting

- Set A1C, blood glucose, and time-in-range goals
- Set lipid goal
- If hypertension is present, establish blood pressure goal
- Weight management and physical activity goals
- Diabetes self-management goals

Therapeutic treatment plans

- Lifestyle management (e.g., registered dietitian nutritionist)
- Pharmacologic therapy: glucose lowering
- Pharmacologic therapy: cardiovascular and kidney disease risk factors
- Weight management with pharmacotherapy or metabolic surgery, as appropriate
- Use of glucose monitoring and insulin delivery devices
- Referral to diabetes education and medical specialists (as needed)

Referrals for initial care management

- Eye care professional for annual dilated eye exam
- Family planning for individuals of childbearing potential
- Registered dietitian nutritionist for medical nutrition therapy
- Diabetes self-management education and support
- Dentist for comprehensive dental and periodontal examination
- Behavioral health professional, if indicated
- Audiology, if indicated
- Social worker and community resources, if indicated
- Rehabilitation medicine or another relevant health care professional for physical and cognitive disability evaluation, if indicated
- Other appropriate health care professionals

Assessment and treatment planning are essential components of initial and all follow-up visits. ASCVD, atherosclerotic cardiovascular disease; MASH, metabolic dysfunction-associated steatohepatitis; MASLD, metabolic dysfunction-associated steatotic liver disease.

Autoimmune Diseases**Recommendations**

4.6 Screen people with type 1 diabetes for autoimmune thyroid disease soon after diagnosis and thereafter at repeated intervals if clinically indicated. **B**

4.7 Adults with type 1 diabetes should be screened for celiac disease in the presence of gastrointestinal symptoms, signs, laboratory manifestations, or clinical suspicion suggestive of celiac disease. **B**

People with type 1 diabetes are at increased risk for other autoimmune diseases, with thyroid disease, celiac disease, and pernicious anemia (vitamin B12 deficiency) being among the most common (17). Other autoimmune conditions associated with type 1 diabetes include autoimmune liver disease, primary

adrenal insufficiency (Addison disease), vitiligo, collagen vascular diseases, and myasthenia gravis (18,19). Type 1 diabetes may also occur with other autoimmune diseases in the context of specific genetic disorders such as polyglandular autoimmune syndromes (20). Given the high prevalence, nonspecific symptoms, and insidious onset of primary hypothyroidism, routine screening for thyroid dysfunction is recommended for all people with type 1 diabetes. Screening for celiac disease should be considered in adults with diabetes with suggestive symptoms (e.g., diarrhea, malabsorption, and abdominal pain) or signs (e.g., osteoporosis, vitamin deficiencies, and iron deficiency anemia) (21). Measurement of vitamin B12 levels (and reflex methylmalonic acid) should be considered for people with type 1 diabetes and peripheral neuropathy or unexplained anemia.

Bone Health**Recommendations**

4.8 Assess fracture risk in older adults with diabetes as a part of routine care in diabetes clinical practice, according to risk factors and comorbidities. **A**

4.9 Monitor bone mineral density using dual-energy X-ray absorptiometry in older adults with diabetes (aged ≥ 65 years) and younger individuals with diabetes and multiple risk factors every 2–3 years (**Table 4.4**). **A**

4.10 Consider the potential adverse impact on skeletal health when selecting pharmacological options to lower glucose levels in people with diabetes. Avoiding medications with a known association with higher fracture risk (e.g., thiazolidinediones and sulfonylureas) is recommended, particularly for those at elevated risk for fractures. **B**

4.11 To reduce the risk of falls and fractures, glycemic management goals should be individualized for people with diabetes at a higher risk of fracture. **C** Prioritize use of glucose-lowering medications that are associated with low risk for hypoglycemia to avoid falls. **B**

4.12 Advise people with diabetes on their intake of calcium (1,000–1,200 mg/day) and vitamin D to ensure it meets the recommended daily allowance for those at risk for fracture, either through their food choices or supplemental means. **B**

4.13a Consider osteoporosis drug therapy in older adults with diabetes who are at increased risk of fracture, including those with low bone mineral density (T-score ≤ -2.5), history of fragility fracture, or elevated Fracture Risk Assessment Tool score ($\geq 3\%$ for hip fracture or $\geq 20\%$ for major osteoporotic fracture). **B**

4.13b Treatment may be considered for adults with diabetes with a T-score between -2.0 and -2.5 in the presence of additional risk factors for fracture. **C**

Determination of fracture risk traditionally has relied on measurements of bone mineral density (BMD) and the World Health Organization–defined T-score of ≤ -2.5 SD. However, it is now established that the consideration of other risk factors improves the categorization of fracture risk

Table 4.3—Highly recommended immunizations for people with diabetes (from the Advisory Committee on Immunization Practices and Centers for Disease Control and Prevention)

Vaccine	Recommended ages	Schedule	GRADE evidence type*	References
COVID-19	All people 6 months of age and older	Current initial vaccination and boosters	3	Centers for Disease Control and Prevention, Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States (305)
Hepatitis B	Adults with diabetes aged <60 years; for adults aged ≥60 years, hepatitis B vaccine may be administered at the discretion of the treating clinician based on the person's likelihood of acquiring hepatitis B infection		1	Sandul et al., Updated Recommendation for Universal Hepatitis B Vaccination in Adults Aged 19–59 Years - United States, 2024 (306)
Influenza	All people with diabetes advised to receive a trivalent influenza vaccine and not to receive live attenuated influenza vaccine	Annual	3	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025 (6,7)
Pneumonia (PPSV23 [Pneumovax])	19–64 years of age, vaccinate with Pneumovax	One dose is recommended for those who previously received PCV13; if PCV15 was used, follow with PPSV23 ≥1 year later; PPSV23 is not indicated after PCV20; adults who received only PPSV23 may receive PCV15 or PCV20 ≥1 year after their last dose	2	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025 (6,7)
	≥65 years of age	One dose is recommended for those who previously received PCV13; if PCV15 was used, follow with PPSV23 ≥1 year later; PPSV23 is not indicated after PCV20; adults who received only PPSV23 may receive PCV15 or PCV20 ≥1 year after their last dose	2	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025 (6,7)
PCV20 or PCV15	Adults 19–64 years of age with an immunocompromising condition (e.g., chronic renal failure), cochlear implant, or cerebrospinal fluid leak Adults 19–64 years of age, immunocompetent ≥65 years of age, immunocompetent, have shared decision-making	One dose of PCV15 or PCV20 is recommended by the Centers for Disease Control and Prevention For those who have never received any pneumococcal vaccine, the Centers for Disease Control and Prevention	2	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19

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Table 4.3—Continued

Vaccine	Recommended ages	Schedule	GRADE evidence type*	References
	discussion with health care professionals	recommends one dose of PCV15 or PCV20 One dose of PCV15 or PCV20; PCSV23 may be given ≥ 8 weeks after PCV15; PPSV23 is not indicated after PCV20		Years or Older - United States, 2025 (6,7)
RSV	Older adults ≥ 60 years of age with diabetes appear to be a risk group	Adults aged ≥ 75 years and those aged ≥ 60 years and at high risk may receive a single dose of an RSV vaccine	1	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025 (6,7)
Tetanus, diphtheria, pertussis (Tdap)	All adults; pregnant individuals should have an extra dose	Booster every 10 years	2	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025 (6,7)
Zoster	≥ 50 years of age	Two-dose Shingrix, even if previously vaccinated	1	Advisory Committee on Immunization Practices Recommended Immunization Schedule for Children and Adolescents Aged 18 Years or Younger - United States, 2025, and Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025 (6,7)

For a comprehensive list of vaccines, refer to the Centers for Disease Control and Prevention web site at cdc.gov/vaccines/. Advisory Committee on Immunization Practices recommendations can be found at cdc.gov/acip/?CDC_AAref_Val=https://cdc.gov/vaccines/acip/recommendations. COVID-19, coronavirus disease 2019; GRADE, Grading of Recommendations Assessment, Development, and Evaluation; PCV13, 13-valent pneumococcal conjugate vaccine; PCV15, 15-valent pneumococcal conjugate vaccine; PCV 20, 20-valent pneumococcal conjugate vaccine; PPSV23, 23-valent pneumococcal polysaccharide vaccine; RSV, respiratory syncytial virus. *Evidence type: 1, randomized controlled trials (RCTs) or overwhelming evidence from observational studies; 2, RCTs with important limitations or exceptionally strong evidence from observational studies; 3, observational studies or RCTs with notable limitations; 4, clinical experience and observations, observational studies with important limitations, or RCTs with several major limitations.

(Table 4.4). There are factors beyond BMD that contribute to bone strength in people with diabetes.

A low-trauma fracture (defined as a fracture occurring from minimal trauma, such as falling from standing height or

less) of the hip, pelvis, vertebra, or forearm in adults aged ≥ 65 years is diagnostic of osteoporosis, regardless of BMD. Such fractures are among the strongest predictors of future fracture risk, especially in the first 1–2 years after a

fracture (22,23). Osteoporotic fractures, especially those of the hip, are associated with significant morbidity, mortality, and societal costs (24). It is estimated that 20% of individuals do not survive to 1 year after hip fracture, while 60% do

Table 4.4—Diagnostic assessment

Individuals who should receive BMD testing

People aged ≥ 65 yearsPostmenopausal women and men aged ≥ 50 years with history of adult-age fracture or with diabetes-specific risk factors:

- Frequent hypoglycemic events
- Diabetes duration >10 years
- Diabetes medications: insulin, thiazolidinediones, sulfonylureas
- A1C $>8\%$
- Peripheral or autonomic neuropathy, retinopathy, nephropathy
- Frequent falls
- Glucocorticoid use: prednisone at doses >2.5 mg per day for ≥ 3 months

not regain their prior functionality, living with permanent disability (25).

Hip fractures in people with diabetes are associated with higher risk of mortality (28% in women and 57% in men), longer recovery, and delayed healing (26) compared with individuals without diabetes.

Epidemiology and Risk Factors

Age-specific fracture risk is significantly increased in people with type 1 or type 2 diabetes in both sexes, with a 34% increase in fracture risk compared with those without diabetes (27).

Type 1 Diabetes. Fracture risk in people with type 1 diabetes is increased by 4.35 times for hip fractures, 1.83 times for upper-limb fractures, and 1.97 times for ankle fractures (28). Fractures occur at young ages, 10–15 years earlier than they do in people without diabetes. Type 1 diabetes is often associated with low bone mass, although BMD underestimates the high risk of fracture observed in young individuals (28). Risk of fracture is increased in people with type 1 diabetes with microvascular complications or neuropathy compared with those without these complications. (26). Moreover, average A1C $>7.9\%$ (risk ratio [RR] 3.57 [CI 1.08–11.78]), duration of diabetes >26 years (RR 7.6 [CI 1.67–34.6]), and family history of fractures (RR 2.64 [CI 1.15–6.09]) have been independently associated with high risk of nonvertebral fractures (29).

Type 2 Diabetes. In people with type 2 diabetes, even with normal or higher BMD, hip fracture risk is increased by 1.79 times, and risk throughout life is 40–70% higher than in it is in individuals without diabetes (27,30,31). According to a meta-analysis that included 15 studies, people with type 2 diabetes had a

35% higher incidence of vertebral fractures, causing increased risk of mortality (hazard ratio [HR] 2.11 [95% CI 1.72–2.59]) (32). Fracture risk is also increased in the upper limbs and ankle. Despite normal to high BMD compared with individuals without type 2 diabetes, the velocity of bone loss with age is greater in individuals with type 2 diabetes, and low BMD remains a risk factor for fractures (33,34).

Glycemic management significantly impacts fracture risk in people with diabetes. A meta-analysis revealed an 8% increased fracture risk per 1% rise in A1C level (RR 1.08 [95% CI 1.03–1.14]) (35). Poor glycemic management (A1C $>9\%$) over 2 years in individuals with type 2 diabetes correlated with a 29% heightened fracture risk (36). Notably, this risk was higher among White individuals than in other racial groups. Hypoglycemia also escalates the risk of fractures at the hip and other skeletal sites (RR 1.52 [95% CI 1.23–1.88]) (35). A Japanese study echoed these findings, showing a fracture risk increase (HR 2.24 [95% CI 1.56–3.21]) with severe hypoglycemia episodes (37).

Longer diabetes duration further elevates fracture risk (38,39). Data indicate individuals who have had type 2 diabetes for >10 years face significantly higher fracture risks. Longer duration of disease may be associated with deposition of advanced glycation end products into the bone matrix, microvascular damage, and a proinflammatory state, which all impair bone cell activity and bone microdamage repair (40). Additionally, high fracture risk is seen in people with CVD, nephropathy, retinopathy, neuropathy, poor physical function, and frequent falls (41–43).

Screening

Most evidence on screening in individuals at risk for fracture is available from

people with type 2 diabetes; fracture risk prediction using BMD in type 1 diabetes has not been extensively studied. Health care professionals should assess fracture history and risk factors in people with diabetes and recommend measurement of BMD if appropriate according to the individual's age and sex.

Type 2 Diabetes. People with type 2 diabetes have 5–10% higher BMD than people without diabetes, although they present with lower bone strength, impaired bone microarchitecture, and accelerated bone loss (33,44–46). Fracture risk was higher in large observational studies in participants with diabetes compared with those without diabetes for a given T-score and age (34). Therefore, a T-score adjustment of -0.5 has been proposed to improve fracture prediction by dual-energy X-ray absorptiometry (DXA). For example, a T-score ≤ -2.0 should be interpreted as equivalent to -2.5 in a person without diabetes. Notably, the Fracture Risk Assessment Tool (FRAX), a medical calculator that predicts 10-year risk of fractures, does not incorporate type 2 diabetes as a risk factor and thus has also been shown to underestimate fracture risk for this population (34). Building on the recognized limitations of FRAX in type 2 diabetes, multiple large cohort studies and meta-analyses consistently demonstrate that type 2 diabetes is an independent risk factor for fracture and that FRAX systematically underestimates fracture risk in this population. This underestimation is observed across diverse ethnic groups, increases with longer diabetes duration, and is particularly pronounced for hip fractures and in older adults with diabetes. Several strategies have been proposed to improve FRAX performance in type 2 diabetes, including 1) using the rheumatoid arthritis input as a surrogate for diabetes, 2) lowering the femoral neck T-score input by 0.5 SD, 3) increasing the age by 10 years, or 4) incorporating trabecular bone score adjustments (47–49). However, recent findings indicate that fracture risk may be similar for a given femoral neck T-score in people with and without diabetes (39).

Fracture risk prediction may be enhanced by use of trabecular bone score (50,51), although such studies are not available for individuals with type 1

diabetes and are based on data from the U.S. or Canada.

In people with type 2 diabetes, BMD should be monitored by DXA scan in older adults (aged ≥ 65 years) in the absence of other comorbidities and in younger individuals (≥ 50 years of age) with bone or diabetes-related risk factors, such as insulin use or diabetes duration >10 years (Table 4.4). Reassessment is recommended every 2–3 years (48), depending on the screening evaluation, antihypertension medication initiation/monitoring, and the presence of additional risk factors, although the evidence on how frequently DXA should be repeated is less robust. According to the European Association for the Study of Obesity (EASO), DXA should be performed every 2 years in individuals undergoing metabolic surgery.

DXA-assisted vertebral fracture assessment is a convenient and low-cost method to assess vertebral fractures, although traditional lateral thoracic/lumbar spine X-ray is still considered the gold standard (52). MRI or computed tomography imaging studies performed for other purposes should be analyzed for presence of vertebral fractures as well as chest X-rays in hospitalized individuals.

Type 1 Diabetes. Because hip fracture risk in type 1 diabetes starts to increase after the age of 50 years, clinicians may consider assessing BMD after the 5th decade of life or earlier in the presence of low-trauma fracture or additional risk factors for bone loss (28). In people with type 1 diabetes, BMD underestimates fracture risk, but studies do not address the extent of underestimation of fracture risk.

According to the International Society for Pediatric and Adolescent Diabetes (ISPAD), regular assessment of bone health using bone densitometry in children and adolescents with type 1 diabetes is still controversial and not recommended, but it may be considered in association with celiac disease (51).

Management

Appropriate glycemic management and minimizing hypoglycemic episodes are crucial for bone health in people with diabetes. Individuals with prolonged disease, microvascular and macrovascular complications, or frequent hypoglycemic episodes face higher fracture risks and fall risks due to factors like

poor vision, neuropathy, sarcopenia, and impaired gait. Health care professionals should advocate for moderate physical activity to enhance muscle health, gait coordination, and balance as part of fracture preventive strategies (42,43,53). For specific recommendations on physical activity for older adults, please refer to section 13, “Older Adults.”

Osteoporosis and fracture prevention are first based on measures applied to the general population. All people with diabetes should receive an adequate daily intake of protein, calcium, and vitamin D and stop smoking (54–56). All individuals should engage in regular physical activity, including aerobic activities and weight-bearing, flexibility, and balance exercise.

Intake of calcium (1,000–1,200 mg) should reflect the age-specific recommendations for the general population and should be obtained through nutrition plan and/or oral supplements (57).

The optimal level of 25-hydroxyvitamin D is a matter of controversy (58), although serum levels 20–30 ng/mL are generally thought to be sufficient (59). For individuals receiving antiresorptive therapy (including bisphosphonate therapy), a 25-hydroxyvitamin D level of >30 ng/mL has been recommended (60,61).

The safe upper limit is also a matter of debate, and there is substantial disagreement over whether to treat to a specified serum level. In the U.S., the recommended daily allowance of vitamin D is 600 IU for people aged 51–70 years and 800 IU for people aged >70 years (59). In clinical practice, this dose of supplement may not be sufficient to reach recommended serum levels of vitamin D, particularly in those at risk for vitamin D deficiency, and therefore supplementation should be individualized. Of note, recent draft recommendations made by the U.S. Preventive Services Task Force caution against supplementation with vitamin D (with or without calcium) for the primary prevention of fractures in community-dwelling postmenopausal women and men aged 60 years or older (62).

Fractures are important outcomes of frailty, a predisability condition that should be mitigated with individualized interventions to prevent falls, maintain mobility, and delay disability (63). In many circumstances, conservative management (calcium, vitamin D, and lifestyle measures) are not enough to reduce fracture risk.

When pharmacological treatment is needed, treatment initiation strategies are the same as those used for the general population. Antihypertension medications reduce bone resorption (bisphosphonates, selective estrogen receptor modulators, and denosumab), stimulate bone formation (teriparatide and abaloparatide), or have dual actions by stimulating bone formation and reducing bone resorption (romosozumab). These agents improve bone density and reduce the risk of vertebral and nonvertebral fractures. Although there are no studies specifically designed for people with diabetes, data on antiresorptive and osteoanabolic agents suggest efficacy in type 2 diabetes is similar to that for individuals without diabetes (64–66). Using individual participant data from randomized trials, antiresorptive therapies reduce fracture risk in people with and without type 2 diabetes for vertebral, hip, and nonvertebral fractures (64). No similar studies of efficacy of antihypertension treatment in people with type 1 diabetes have been published.

Primary Prevention of Fragility Fractures in People With Diabetes. In the general population, it is recommended that people with diabetes with a T-score ≤ -2.5 or an elevated FRAX score ($\geq 3\%$ for hip fracture or $\geq 20\%$ for major osteoporotic fracture) should be considered for pharmacological treatment with first-line drug therapy (alendronate, risedronate, ibandronate, and zoledronate or denosumab). In type 2 diabetes, since T-score underestimates fracture risk (as discussed above), pharmacological treatment should be strongly considered for adults with diabetes with a T-score between -2.0 and -2.5 in the presence of additional risk factors for fracture such as falls and elevated FRAX score.

The risk of hypocalcemia associated with denosumab use in individuals with CKD is markedly increased in advanced CKD (estimated glomerular filtration rate <30 mL/min/1.73 m²) and is highest in dialysis-dependent individuals. This risk must be proactively managed with pretreatment assessment, supplementation, close monitoring, and specialist consultation. Data from a phase 3 trial, Future Revascularization Evaluation in Patients With Diabetes Mellitus: Optimal Management of Multivessel Disease (FREEDOM), and its 10-year extension have shown that people with diabetes treated with denosumab

experienced positive effects on fasting glucose (67) and significant improvements in BMD and lower vertebral fracture risk (49). However, according to a post hoc subgroup analysis, a higher risk of nonvertebral fractures was observed in people with diabetes treated with denosumab (49). Self-management abilities of the person with diabetes should be considered in medication selection because inability to maintain the every 6-month denosumab injection schedule can lead to rebound bone loss and spontaneous vertebral fractures, careful individual selection and counseling are important prior to drug initiation. Bisphosphonate therapy (oral or intravenous) may be more appropriate in individuals with poor medication-taking behavior or gaps in access to medical care. As diabetes is a low-bone-turnover condition, very-high-risk individuals should be considered for osteoanabolic therapy (teriparatide, abaloparatide, romosozumab). Teriparatide showed positive effects on BMD and fracture risk in people with diabetes (65,66).

Romosozumab received FDA approval only for postmenopausal women with a box warning because it may increase risk of myocardial infarction, stroke, or cardiovascular death. Although real-world data suggests that romosozumab may not lead to an increase in cardiac events (68), it should not be prescribed in individuals who experienced a myocardial infarction or a stroke within the past year (69,70). Osteoanabolic treatments must be followed by antiresorptives to preserve BMD gains. Without consolidation, BMD declines within a year as increased bone turnover leads to rapid loss of newly formed bone.

Secondary Prevention of Fragility Fractures.

The risk of subsequent low-trauma fracture (hip, spine, nonvertebral site) is high, especially in the first 1–2 years after a fracture. Antiosteoporosis treatment reduces the risk of fracture in older individuals with prior hip or vertebral fracture.

As in the general population, people with diabetes who experience fragility fracture should 1) be given the diagnosis of osteoporosis regardless of bone density data and 2) receive the appropriate work-up and therapy to prevent future fractures (71). Individuals on long-term treatment with antiosteoporosis medications, with multiple fragility fractures, or with multiple comorbidities should be referred to a bone metabolic specialist. In these more complicated cases, a bone specialist may

choose to initiate an osteoanabolic agent to optimize bone formation and reduce immediate fracture risk (72). It is strongly recommended that all individuals with a fragility fracture be started on antiosteoporosis therapy and adequate calcium and vitamin D supplementation (if required) as soon as possible. In the appropriate individual, therapy may even be initiated during an inpatient stay to reduce care delays (71).

Glucose-Lowering Medications and Bone Health

Care plans for type 2 diabetes treatment should consider individual fracture risk and the potential effect of medications on bone metabolism. Metformin has a safe bone health profile (73,74). Studies have reported increased fracture incidences in women using thiazolidinediones (TZD), with the risk doubling with 1–2 years of TZD use compared with placebo or other glucose-lowering medications (HR 2.23 [95% CI 1.65–3.01]) (75,76). According to the Action to Control Cardiovascular Risk in Diabetes (ACCORD) study, reduced risk is noted in women who had discontinued TZD use for 1–2 years (HR 0.57 [95% CI 0.35–0.92]) or >2 years (HR 0.42 [95% CI 0.24–0.74]) compared with current users (77). Furthermore, individuals with type 2 diabetes on sulfonylureas exhibit a heightened fracture risk (RR 1.30 [95% CI 1.18–1.43]) (78) because of accelerated bone loss (46) and a higher rate of hypoglycemic events, which can trigger falls and fractures (79). Dipeptidyl peptidase 4 inhibitors have been used in clinical practice for more than 15 years, and both clinical trials and postmarketing data suggest a neutral impact on bone health (80,81). Glucagon-like peptide 1 receptor agonists (GLP-1 RAs) at low doses are likely neutral to beneficial with respect to BMD and fracture risk (82), though data are lacking for higher doses in which weight loss may be higher and more rapid with potential counteractive skeletal risk. Use of sodium–glucose cotransporter 2 (SGLT2) inhibitors has raised some concerns. The Canagliflozin Cardiovascular Assessment Study (CANVAS) showed that the proportion of subjects with fracture was higher in the canagliflozin groups than the non-canagliflozin groups (2.7% vs. 1.9%, respectively). Further analyses from the same trial and from the Canagliflozin and Renal Events in Diabetes with Established Nephropathy Clinical Evaluation (CRENDENCE) study found a neutral effect on fracture risk

(83–86). Although few data are available, use of empagliflozin, ertugliflozin, or dapagliflozin has not been associated with negative effects on bone health (85–87). Use of insulin has been shown to be associated with a doubling of the risk of hip fractures (79), likely because of higher risk of hypoglycemia, longer duration of the disease, and comorbidities that may contribute to diminished bone strength.

In conclusion, glucose-lowering medications with a good bone safety profile are preferred. This is especially true in older adults, in people with longer duration of diabetes, or in people with complications. Aggressive therapeutic approaches should be avoided in those who are frail and in older adults to prevent hypoglycemic events and falls.

Cancer

Diabetes is associated with increased risk of cancers of the liver, pancreas, endometrium, colon and rectum, breast, and bladder (88). The association may result from shared risk factors between type 2 diabetes and cancer (older age, obesity, and physical inactivity) but may also be due to diabetes-related factors (89), such as underlying disease physiology or diabetes treatments, although evidence for these links is scarce. People with diabetes should be encouraged to undergo recommended age- and sex-appropriate cancer screenings, coordinated with their primary health care professional, and to reduce their modifiable cancer risk factors (obesity, physical inactivity, alcohol use, and smoking). New onset of atypical diabetes (lean body habitus and negative family history) in a middle-aged or older person may precede the diagnosis of pancreatic adenocarcinoma (90). Additionally, in a nationwide cancer registry in New Zealand, postpancreatitis diabetes mellitus was associated with significantly higher risk (2.4-fold) of pancreatic cancer compared with pancreatitis after type 2 diabetes (91). However, in the absence of other symptoms (e.g., weight loss and abdominal pain), routine screening for pancreatic cancer is not currently recommended. Metformin and sulfonylureas may have anticancer properties. Data for pioglitazone are mixed, with a previous concern for bladder cancer association. Recommendations cannot be made at this time (92–94). Thus far, the use of GLP-1 RAs has not been shown to be

associated with the incidence of thyroid cancer, pancreatic cancer, or any other type of cancer in humans (95,96). For information on glycemic management in the context of cancer treatment, see section 9, "Pharmacologic Approaches to Glycemic Treatment."

Cognitive Impairment/Dementia

Recommendation

4.14 In the presence of cognitive impairment, diabetes treatment plans should be simplified as much as possible and tailored to minimize the risk of hypoglycemia. **B**

Diabetes is associated with a significantly increased risk and rate of cognitive decline and an increased risk of dementia (97). A meta-analysis of 122 prospective observational studies found that individuals with diabetes had a 43% higher risk of all types of dementia, a 43% higher risk of Alzheimer dementia, and a 91% higher risk of vascular dementia compared with individuals without diabetes (97). In a 15-year prospective study of community-dwelling people >60 years of age, the presence of diabetes at baseline significantly increased the age- and sex-adjusted incidence of all-cause dementia, Alzheimer dementia, and vascular dementia compared with rates in those with normal glucose tolerance (98). The reverse is also true: people with Alzheimer dementia are more likely to develop diabetes than people without Alzheimer dementia, possibly linking critical metabolic processes that contribute to neurodegeneration (99). A new clinical entity of diabetes-related dementia is being recognized as distinct from Alzheimer dementia or vascular dementia. It is characterized by slow progression of dementia, absence of typical neuroimaging findings seen in Alzheimer or vascular dementia, old age, high A1C levels, long duration of diabetes, high frequency of insulin use, frailty, and sarcopenia or dynapenia (100). See section 13, "Older Adults," for a more detailed discussion regarding assessment of cognitive impairment.

Glycemic Status and Cognition

In individuals with diabetes, higher A1C level is associated with lower cognitive function (101). A meta-analysis of randomized trials found that intensive

glycemic management, compared with higher A1C goals, was associated with a slightly lower rate of cognitive decline. However, these findings were driven by an older study with an A1C goal of <7.0% in the intensive treatment arm. Analyses within the ACCORD, Action in Diabetes and Vascular Disease: Preterax and Diamicon MR Controlled Evaluation (ADVANCE), and Veterans Affairs Diabetes Trial (VADT) studies found that intensive glycemic management (A1C goal of <6.0–6.5%) resulted in no differences in cognitive outcomes compared with standard control (102,103).

Results from the China Health and Retirement Study (CHARLS) report that in adults aged >45 years with type 2 diabetes, excessively low and high A1C levels (>9%) were associated with increased risk of cognitive impairment (104). Therefore, intensive glycemic management should not be advised for the improvement of cognitive function in individuals with type 2 diabetes. Additionally, people with type 2 diabetes and dementia are at heightened risk for experiencing hyperglycemic crises (diabetic ketoacidosis and hyperglycemic hyperosmolar state) compared with people without dementia (105), underscoring the importance of supporting and simplifying diabetes management for individuals experiencing cognitive decline and diminished capacity for self-care. In addition, these individuals have increased difficulty with complex treatment and monitoring plans and are at risk for frailty, hypoglycemia, and disability (106).

In type 2 diabetes, severe hypoglycemia is associated with reduced cognitive function, and those with poor cognitive function have more severe or repeated episodes of hypoglycemia. Multiple observational studies of adults with diabetes have found an association between severe hypoglycemic episodes and cognitive decline or incident dementia (107–110). Decreased cognitive function also increases the risk for severe hypoglycemia, likely through impaired ability to recognize and respond appropriately to hypoglycemic symptoms (107,111,112). Additionally, long-term follow up of Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complications (DCCT/EDIC) showed recurrent severe hypoglycemia was associated with the highest risk of long-term psychomotor and mental function decline (113). Simplifying

or deintensifying glycemic therapy and/or liberalizing A1C goals may prevent hypoglycemia in individuals with cognitive dysfunction. See section 13, "Older Adults," for more detailed discussion of hypoglycemia in older people with type 1 and type 2 diabetes.

Dental Care

Recommendations

4.15 People with diabetes should be referred for a dental exam at least once per year. **E**

4.16 Coordinate efforts between the medical and dental teams to appropriately adjust glucose-lowering medication and treatment plans prior to and in the postdental procedure period as needed. **B**

Periodontal disease is more severe, and may be more prevalent, in people with diabetes than in those without and has been associated with higher A1C levels (114,115). Longitudinal studies suggest that people with periodontal disease have higher rates of incident diabetes, suggesting a bidirectional relationship between diabetes and periodontal disease. Current evidence suggests that periodontal disease adversely affects diabetes outcomes, and periodontal treatment using subgingival instrumentation may improve glycemic outcomes (116,117).

The proportion of U.S. adults with diabetes seeing a dentist in the past year was significantly lower than that of individuals without diabetes (56.7% vs. 65.4%). The majority of clinicians do not discuss oral health with individuals (118). Increased engagement in toothbrushing has been shown to be associated with improved oral health and better glycemic management in people with type 2 diabetes (119). Dental health professionals should be included in the diabetes care team when possible, using a collaborative interprofessional care approach (120). Early detection of oral health problems by clinicians may be helpful to promote prompt referral to dental care and mitigate the expensive and extensive procedures needed to treat advanced oral disease (121,122). Clinical assessment of people with diabetes should include a dental history and an oral examination, and dental professionals should be informed about key aspects of the person's

health and diabetes treatment plan, including glycemic goals, medications, and comorbid conditions (121,122). It is important for dental professionals to know when people with diabetes have high A1C levels, as this population may have lower oral healing capacity (123,124). Hepatic, cardiovascular, kidney, and pulmonary conditions should also be known by dental professionals to assist in appropriate dosing of antibiotics and other medications. Coordination between dental professionals and the diabetes care team will be especially important for people treated with insulin, sulfonylureas, or meglitinides who are at risk for hypoglycemia during dental procedures, especially if fasting. Hypoglycemia can be mitigated by medication adjustment, blood glucose monitoring before and during the procedure, and treatment of hypoglycemia if appropriate. Therefore, dental professionals caring for people with diabetes should have access to blood glucose monitors during procedures as well as carbohydrates and glucagon to treat any hypoglycemia that occurs.

Disability

Recommendation

4.17 Assess for disability at the initial visit and for decline in function at each subsequent visit in people with diabetes. If a disability is impacting functional ability or capacity to manage their diabetes, a referral should be made to an appropriate health care professional specializing in disability (e.g., physical medicine and rehabilitation specialist, physical therapist, occupational therapist, or speech-language pathologist). **C**

A disability is defined as a physical or mental impairment that substantially limits one or more major life activities of an individual (125,126). Activities of daily living (ADLs) and instrumental activities of daily living (IADLs) comprise basic and complex life care tasks, respectively. The capacity to accomplish such tasks serves as an important measure of function. Diabetes is associated with an increase in the risk of work and physical disability, with estimates of 50–80% increased risk of disability for people with diabetes compared with people without diabetes (127). Reviews have shown that lower-body functional

limitation was the most prevalent disability (47–84%) among people with diabetes (128,129). In a systematic review and meta-analysis, the presence of diabetes increased the risk of mobility disability (15 studies; odds ratio [OR] 1.71 [95% CI 1.53–1.91]; RR 1.51 [95% CI 1.38–1.64]), of IADL disability (10 studies; OR 1.65 [95% CI 1.55–1.74]), and of ADL disability (16 studies; OR 1.82 [95% CI 1.63–2.04]; RR 1.82 [95% CI 1.40–2.36]) (127). In a meta-analysis, diabetes was shown to be associated with an increased risk of falls in older adults; falls occurred in 25% of individuals with diabetes compared with 18.2% of those without diabetes (130). The mechanisms underlying disability are multifactorial and include obesity, coronary artery disease, stroke, lower extremity complications, and physiological factors such as hyperglycemia, sarcopenia, inflammation, and insulin resistance (131).

Diabetic peripheral neuropathy (DPN) is a common complication of both type 1 and 2 diabetes and may cause impaired postural balance and gait kinematics (132), leading to functional disability. DPN can be found in up to half of people with type 1 or type 2 diabetes, resulting in physical disability, and neuropathic pain, resulting in a diminished quality of life (133). Glycemic management prevents DPN development in type 1 diabetes; in contrast, glycemic management has modest or no benefit in individuals with type 2 diabetes, possibly due to the combined effect of coexisting comorbidities (133). People with lower-extremity involvement due to DPN have 3 times more risk of restricted mobility, resulting in them experiencing more physical dysfunctions and impairments than people who have diabetes but not neuropathy (134). Furthermore, DPN may progress to nontraumatic lower-limb amputation, which significantly impacts quality of life (135).

In addition to complications of diabetes from microvascular conditions such as CKD, retinopathy, autonomic neuropathy, and peripheral neuropathy, it is important to recognize the disabilities caused by macrovascular complications of diabetes. These macrovascular complications, which include coronary heart disease, stroke, and peripheral arterial disease, can lead to further impairments (128).

An assessment of disability should be performed to evaluate individuals' ability to perform ADLs and IADLs, ensuring they can manage basic self-care and more complex tasks necessary for specific living situations. Referrals may be made as necessary to appropriate health care professionals specializing in disability (e.g., physical medicine and rehabilitation physician, physical therapist, occupational therapist, or speech-language pathologist) (127,136,137). Customized rehabilitation interventions for individuals with a disability from diabetes can recover function, allowing for safe physical activity (138), and improve quality of life (139). Additionally, frailty is commonly associated with diabetes, with progression to disability, morbidity, and mortality in older adults. People with diabetes as well as frailty or disability may be sedentary and contend with comorbid conditions such as hypoglycemia, sarcopenia, falls, nutritional deficiencies, obesity, polypharmacy, and cognitive dysfunction. A thorough medical evaluation is imperative to identify the best approaches to preventative and therapeutic interventions for frailty and diabetes management (140).

A psychosocial assessment should be conducted to screen for behavioral health conditions like depression and anxiety and to understand the individual's social support and coping mechanisms. Functional capacity evaluations, involving tests for physical endurance and strength, are used to gauge the ability of the person with diabetes to work and carry out daily activities. Additionally, standardized disability questionnaires and scales, such as the Diabetes Distress Scale (DDS) and the World Health Organization Disability Assessment Schedule (WHODAS 2.0), are employed to measure the emotional burden of diabetes and overall disability (141,142). These suggested structured assessments are particularly relevant if individuals have fallen, had emergency department visits, missed appointments, made significant errors in the treatment plan, or exhibit apathy and depressed mood.

Moreover, when treating people with an acquired disability from diabetes, it is vital to consider social determinants of health, race and ethnicity, and socioeconomic status (143). Rates of diabetes-related major amputations are higher in individuals who are from racial and ethnic minoritized groups (144), live in rural areas, and are from regions with the

lowest socioeconomic levels (145). Addressing the complex challenges faced by individuals with acquired disabilities from diabetes requires a multifaceted approach involving solutions from both within and outside the health care system. By focusing on social determinants of health, health care professionals can develop appropriate interventions, provide advocacy, and establish support systems that cater to the specific needs of this population. See section 1, "Improving Care and Promoting Health in Populations."

Hepatitis C

Infection with hepatitis C virus (HCV) is associated with a higher prevalence of type 2 diabetes, which is present in up to one-third of individuals with chronic HCV infection. HCV may impair glucose metabolism by several mechanisms, including directly via viral proteins and indirectly by altering proinflammatory cytokine levels (146). The use of newer direct-acting antiviral drugs produces a sustained virological response (cure) in nearly all cases and has been reported to improve glucose metabolism in individuals with diabetes (147). A meta-analysis of mostly observational studies found a mean reduction in A1C levels of 0.45% (95% CI −0.60 to −0.30) and reduced requirement for glucose-lowering medication use following successful eradication of HCV infection (148).

Sexual Health in Men

Recommendations

4.18 In men with diabetes or prediabetes, screen for erectile dysfunction (ED), particularly in those with high cardiovascular risk, retinopathy, cardiovascular disease, chronic kidney disease, peripheral or autonomic neuropathy, longer duration of diabetes, depression, and hypogonadism and in those who are not meeting glycemic goals. **B**

4.19 In men with diabetes or prediabetes, inquire about sexual health (e.g., low libido and ED). If symptoms and/or signs of hypogonadism are detected, screen with a morning serum total testosterone level. **B**

The most common sexual dysfunction in men is erectile dysfunction (ED), with an estimated prevalence of 52.5% in men

with diabetes (149). The best predictors of ED are age (>40 years), CVD, diabetes, hypertension, obesity, dyslipidemia, metabolic syndrome, hypogonadism, smoking, depression, and use of medications such as antidepressants and opioids (150). Because diabetes, poor nutrition, obesity, lack of exercise, and CVD are often interrelated, it may be challenging to identify the primary risk factor (151), although the most likely primary underlying risk factor is vascular disease (151).

Men with diabetes are at increased risk for both CVD and ED, and ED is a predictor of cardiovascular events in men with diabetes (152,153) as well as in men without diabetes. The significant factors associated with ED in men with diabetes are age, peripheral or autonomic neuropathy, presence of microvascular disease including retinopathy, CVD, duration of diabetes, poor glycemic management, hypogonadism, and diuretic therapy (154). Physical activity may be protective. Men with diabetes and ED report a significant decline in quality-of-life measures and an increase in depressive symptoms (155), and depression is a well-recognized risk factor for ED. Given the bidirectional relationship between ED and depression, treatment of either one can result in improvement in the other condition. CKD is also a risk factor for CVD and ED, with prevalence rates of ED >75% in men on hemodialysis (156).

Awareness and identification of these characteristics, factors, and behaviors can guide clinicians in early screening, treatment, prevention, and counseling in all men with diabetes and particularly those at higher risk for ED (154). Given the evidence that ED is strongly associated with diabetes and CVD, men with ED should be evaluated and managed for cardiovascular and endocrine risk factors. Glycemic assessment in men not previously diagnosed with diabetes, lipid profile, and morning total and free testosterone should be considered mandatory in all men newly presenting with ED (157).

In a recent meta-analysis, testosterone was superior to placebo in improving erectile function in men with testosterone deficiency; however, the magnitude of the effect was lower in the presence of diabetes and obesity (158).

Meta-analyses show that all phosphodiesterase type 5 inhibitors (PDE5Is) are superior to placebo in treating ED, lower dosages had effects comparable with

those of higher dosages, and various PDE5Is show comparable efficacy (151). PDE5Is are associated with an increased risk of headaches, flushing, and dyspepsia (151). First-line therapy for ED in men with diabetes is PDE5Is, but men with diabetes may be less responsive than men without diabetes (149). Strategies to improve response to PDE5Is include daily therapy and optimization of comorbidities. In men with diabetes not responding to PDEIs, other potentially effective treatments may include intracavernosal injections, intraurethral prostaglandin, vacuum erection devices, and penile prosthetic surgery (149).

Mean levels of testosterone are lower in men with diabetes than in age-matched men without diabetes, but obesity is a major confounder (159,160). Testosterone replacement in men with symptomatic hypogonadism may have benefits, including improved sexual function, well-being, muscle mass and strength, and bone density (161). In men with diabetes who have symptoms or signs of low testosterone (hypogonadism), a morning total testosterone level should be measured using an accurate and reliable assay (162). In men who have total testosterone levels close to the lower limit, it is reasonable to determine free testosterone concentrations either directly from equilibrium dialysis assays or by calculations that use total testosterone, sex hormone binding globulin, and albumin concentrations (162). Further tests (such as prolactin, luteinizing hormone and follicle-stimulating hormone levels) may be needed to further evaluate the individual. Testosterone replacement in older men with hypogonadism has been associated with increased coronary artery plaque volume, with no conclusive evidence that testosterone supplementation is associated with increased cardiovascular risk in all men with hypogonadism (162,163). Furthermore, ED is also common in people with diabetes (164), and it is reasonable to measure and correct testosterone levels close to the lower limit to address the libido component that contributes to erectile difficulties (151).

Sexual Health in Women

Recommendations

4.20 In women with diabetes or prediabetes, inquire about sexual health by

screening for desire (libido), arousal, and orgasm difficulties, particularly in those who experience depression and/or anxiety and those with recurrent urinary tract infections. **B**

4.21 In postmenopausal women with diabetes or prediabetes, screen for symptoms and/or signs of genitourinary syndrome of menopause, including vaginal dryness and dyspareunia. **B**

Female sexual dysfunction (FSD) is common in women with diabetes. In an epidemiologic cross-sectional study of community-residing middle-aged and older adults (57–85 years), women with diagnosed diabetes were less likely than men with diagnosed diabetes (adjusted OR 0.28 [95% CI 0.16–0.49]) and women without diabetes (0.63 [0.45–0.87]) to be sexually active (165). Older women with diabetes are as likely as men to have sexual problems but are significantly less likely to have discussed sex with a physician (165).

While studies showing the association between diabetes and FSD are less conclusive than those in men, most have reported a higher prevalence of FSD in women with diabetes compared with women without diabetes (166), and a meta-analysis found that FSD is more common in women with type 1 and type 2 diabetes (OR 2.27 and 2.49, respectively) than in women without diabetes (167).

A wide range of prevalence rates of sexual dysfunctions has been reported in women with diabetes. In women with type 1 diabetes, 16–85% (vs. 0–66% in women without diabetes) report problems with desire, 11–76% (vs. 0–41%) report problems with arousal, and 9–66% (vs. 0–39%) report problems with orgasm; 9–57% (vs. 0–28%) report problems with lubrication, and 7–61% (vs. 5–39%) report problems with pain. In women with type 2 diabetes, 70–82% (vs. 10–66% in women without diabetes) report problems with desire, 54–68% (vs. 3–41%) report problems with arousal, and 33–84% (vs. 2–39%) report problems with orgasm; 33–66% (vs. 4–28%) report problems with lubrication, and 33–46% (vs. 8–39%) report problems with pain (168).

Medical comorbidities that are risk factors for FSD include hypertension, obesity,

metabolic syndrome, smoking, and hyperlipidemia. Clinical factors for consideration include longer duration of diabetic retinopathy and neuropathy and individuals not meeting glycemic goals. The prevalence of FSD in women with kidney disease is 74% (169).

In women with diabetes, social and psychological components play a major role in FSD. Depression, anxiety, and emotional adjustments to diabetes have been found to be associated with sexual dysfunction in women with diabetes. A study from Norway reported that women with type 1 diabetes with scores on the Female Sexual Function Index (FSFI) (a validated instrument) indicating sexual dysfunction were more likely than women without sexual dysfunction to have diabetes distress, depression, and menopausal symptoms. They were also older and more likely to be single and postmenopausal (170). Another study also showed that women with sexual dysfunction were significantly more likely to report impaired well-being, elevated diabetes distress, poor adjustment to diabetes, and moderate to severe anxiety than women without sexual dysfunction (168).

In a qualitative study exploring the experiences of sexual health and sexual challenges, women with type 1 diabetes reported that diabetes affected their relationship, including sex life, and had an impact on their partner. Challenges included reduced sexual desire, decline in frequency, less spontaneous desire resulting in lack of initiation, and physical challenges such as pain, vaginal dryness, and impaired sensitivity. Several women explained that vaginal dryness was an obstacle during sexual intercourse, leading to pain or even refraining from sexual activity. Sexual challenges were perceived to become a source of disappointment to the partners and consequential guilt for the women. Women also reported fear of hypoglycemia during sex, and some reported trying to maintain mild hyperglycemia. Technology devices, such as glucose monitors and insulin pumps, could be perceived as both a physical and mental obstacle during sexual activity (171).

Many women with diabetes would like their health care professional to initiate a discussion on how diabetes is affecting their sex life (172). Women with type 1 diabetes almost unanimously

endorsed that sexual health should be addressed, that they would find it a relief that they were not alone, that they should be provided with information when they are young, and that it would be difficult to address the topic themselves (171). Unfortunately, many health care professionals do not actively discuss sexual functioning in consultations, meaning that when the topic is discussed it is mostly the person with diabetes who initiates the conversation (165). This leads to a marked underdiagnosis and undertreatment of sexual dysfunctions in people with diabetes.

While no specific guidelines are available for the treatment of FSD in this population, women with type 1 or type 2 diabetes should be encouraged to engage in lifestyle interventions and, in the absence of contraindications, may benefit from already-approved treatments for FSD (173). The Look AHEAD (Action for Health in Diabetes) study on intervention demonstrated statistical improvements in the FSFI total score and all domains of sexual dysfunction (174). Lifestyle factors that enhance desire and sexual function include nutrition (such as the Mediterranean eating pattern), exercise (such as walking), and smoking cessation. Other interventions include improving glycemic management and prevention of diabetes complications; diagnosis and treatment of menopausal symptoms with hormonal therapies; addressing vaginal dryness and dyspareunia as well as urinary tract and mycotic genital infections; screening and addressing depression, anxiety, diabetes distress, and related psychosocial issues; and considering FDA-approved centrally acting medications for hypoactive sexual desire disorder, including flibanserin and bremelanotide.

Metabolic Dysfunction–Associated Steatotic Liver Disease and Metabolic Dysfunction–Associated Steatohepatitis

Screening

Recommendations

4.22a Screen adults with type 2 diabetes or with prediabetes, particularly those with obesity or other cardiometabolic risk factors or established cardiovascular disease, for their risk of having or developing cirrhosis related to metabolic dysfunction–

associated steatohepatitis (MASH) using a calculated fibrosis-4 index (FIB-4) (derived from age, ALT, AST, and platelets [mdcalc.com/calc/2200/fibrosis4-fib-4-index-liver-fibrosis]), even if they have normal liver enzymes. **B**

4.22b Adults with diabetes or prediabetes with persistently elevated plasma aminotransferase levels for >6 months and low FIB-4 should be evaluated for other causes of liver disease. **B**

4.23 Adults with type 2 diabetes or prediabetes with a FIB-4 ≥ 1.3 should have additional risk stratification by liver stiffness measurement with transient elastography, or, if unavailable, the enhanced liver fibrosis (ELF) test. **B**

4.24 Refer adults with type 2 diabetes or prediabetes at higher risk for significant liver fibrosis (i.e., as indicated by FIB-4, liver stiffness measurement, or ELF) to a gastroenterologist or hepatologist for further evaluation and management. **B**

Metabolic dysfunction–associated steatotic liver disease (MASLD) has replaced the term nonalcoholic fatty liver disease (NAFLD) to identify steatotic liver disease. The definition includes the presence of steatotic liver disease and at least one cardiometabolic risk factor associated with insulin resistance (e.g., prediabetes, diabetes, atherogenic dyslipidemia, or hypertension) without other identifiable causes of steatosis (175). This is in the absence of ongoing or recent consumption of significant amounts of alcohol (defined as ingestion of >21 standard drinks per week in men and >14 standard drinks per week in women over a 2-year period preceding evaluation) or other secondary causes of hepatic steatosis (176). It is estimated that in adults in the U.S., the prevalence of MASLD is >70% of people with type 2 diabetes (177–179). This is consistent with studies from other countries (180,181). The new definition of MASLD aims to remove potential stigma from the term “fatty” when referring to steatosis, highlights the role of prediabetes and type 2 diabetes in MASLD, and provides a positive diagnosis by using cardiometabolic risk factors as surrogates for insulin resistance, the main driver for the development of steatosis. The new definition correlates well with the past definition

of MASLD for people with prediabetes or type 2 diabetes (who already have, by definition, one cardiometabolic risk factor). A separate category outside of MASLD, named metabolic dysfunction and alcoholic liver disease, was created for circumstances in which alcohol intake is greater than that allowed for MASLD but less than that attributed to alcoholic liver disease. More research is needed to better characterize the predictive value for metabolic dysfunction–associated steatohepatitis (MASH) of different cardiometabolic risk factors and the natural history of metabolic dysfunction and alcoholic liver disease or steatosis in young adults without cardiometabolic risk factors.

Diabetes is a major risk factor for developing MASH (formerly nonalcoholic steatohepatitis, or NASH) and worse liver outcomes (180,181). MASH is defined histologically as having $\geq 5\%$ hepatic steatosis with inflammation and hepatocyte injury (hepatocyte ballooning), with or without evidence of liver fibrosis (176). Steatohepatitis is estimated to affect more than half of people with type 2 diabetes with MASLD (182,183). Fibrosis stages are classified histologically as the following: F0, no fibrosis; F1, mild; F2, moderate (significant); F3, severe (advanced); and F4, cirrhosis. In the U.S., between 12 and 20% of people with type 2 diabetes have “at-risk” MASH (i.e., steatohepatitis with clinically significant fibrosis [$\geq F2$] and at risk for cirrhosis) (177,178,182). A similar or higher prevalence has been observed worldwide (180,181,183). People with type 2 diabetes and at-risk MASH are at an increased risk of future cirrhosis, hepatocellular carcinoma (HCC) (184,185), and liver transplantation (186). The prevalence of MASLD in people with type 1 diabetes is $\sim 20\%$ and is driven by obesity, which is becoming more common in this population (187), with a large variability across studies using different steatosis measurement methods (188). The prevalence of liver steatosis in a population with type 1 diabetes by MRI (i.e., the gold standard) with low prevalence of obesity was only 8.8% compared with 68% in people with type 2 diabetes (189). The prevalence of clinically significant fibrosis ($\geq F2$) is estimated to be $\sim 5\%$ (190), which is much lower than the prevalence in type 2 diabetes (177,178,182). Therefore, screening for fibrosis in people

with type 1 diabetes should only be considered in the presence of additional risk factors for MASLD, such as obesity, incidental hepatic steatosis on imaging, or elevated plasma aminotransferases.

Clinicians underestimate the prevalence of at-risk MASH and do not consistently implement appropriate screening strategies in people with prediabetes or type 2 diabetes, thus missing a chance to establish an early diagnosis. This pattern of underdiagnosis is compounded by sparse referral to specialists and inadequate prescription of medications with potential efficacy in MASH (191,192). The goal of screening for MASLD is to identify people with at-risk MASH to prevent future cirrhosis, HCC, liver transplantation, and all-cause mortality (193–196). This risk is higher in people who have central obesity and cardiometabolic risk factors or insulin resistance, are >50 years of age, and/or have persistently elevated plasma aminotransferases (AST and/or ALT >30 units/L for >6 months) (178,197). Some genetic variants that alter hepatocyte triglyceride metabolism may also increase the risk of MASH progression and cirrhosis (198,199), amplifying the impact of obesity, but the role of genetic testing in clinical practice remains to be established. Individuals with MASLD also are at a greater risk of developing extrahepatic cancer (185), type 2 diabetes (194), and CVD (200,201). Emerging evidence suggests that MASLD increases the risk of CKD in people with type 2 diabetes, particularly when liver fibrosis is present (202,203), although the association of MASLD with diabetic retinopathy is less clear (204).

The fibrosis-4 index (FIB-4) is the most cost-effective strategy for the initial screening of people with prediabetes and cardiometabolic risk factors or with type 2 diabetes for at-risk MASH in primary care and diabetes clinical settings (181,192,197,205–208). The diagnostic algorithm for the screening and liver fibrosis risk stratification of people with prediabetes or type 2 diabetes is shown in **Fig. 4.2**. A screening strategy relying on elevated plasma aminotransferases >40 units/L would miss most individuals with MASH in these settings, as at-risk MASH with clinically significant fibrosis ($\geq F2$) is frequently observed with plasma aminotransferases below the commonly used cutoff of 40 units/L (177–179,182,209,210). The American College of Gastroenterology considers

the upper limit of normal ALT levels to be 29–33 units/L for male individuals and 19–25 units/L for female individuals (211), as higher levels are associated with increased liver-related mortality. The FIB-4 estimates the risk of hepatic cirrhosis and is calculated from the computation of age, plasma aminotransferases (AST and ALT), and platelet count (mdcalc.com/calc/2200/fibrosis-4-fib-4-index-liver-fibrosis). A value of <1.3 is considered low risk of having advanced fibrosis (F3–F4) and for developing adverse liver outcomes, while ≥ 1.3 is considered as having a higher probability of at-risk MASH clinically significant fibrosis ($\geq F2$) and increased risk of adverse liver outcomes. A value of >2.67 confers a high risk of having advanced fibrosis (F3–F4), and referral to the liver specialist is warranted without additional testing. FIB-4 predicts changes over time in hepatic fibrosis (212,213) and allows risk stratification of individuals in terms of future liver-related morbidity and mortality (214). FIB-4 has reasonable sensitivity but low specificity; hence, a negative result may rule out fibrosis while a positive result requires confirmatory

testing (213,215,216). Therefore, a FIB-4 score <1.3 should be taken as a general guidance of having a lower risk of MASH with advanced fibrosis but does not replace clinical judgment. Its low cost, simplicity, and good specificity make it the initial test of choice (Fig. 4.2). FIB-4 has not been validated in pediatric populations or in adults aged <35 years. In individuals with diabetes ≥ 65 years of age, higher cutoffs for FIB-4 have been recommended (1.9–2.0 rather than ≥ 1.3) (217). In some individuals with at-risk MASH, FIB-4 score may be <1.3 , especially in those with obesity and type 2 diabetes. Additional testing may be required if the FIB-4 score is between 1.0 and 1.3 in individuals with type 2 diabetes with multiple cardiometabolic risk factors.

In individuals with a FIB-4 ≥ 1.3 , there is need for additional risk stratification with a liver stiffness measurement (LSM) by transient elastography (Fig. 4.2). Use of a second nonproprietary diagnostic panel is not recommended (e.g., MASLD fibrosis score and others), as they generally do not perform better than FIB-4 (176,179,215). Transient elastography (LSM) is the best-validated imaging technique for

fibrosis risk stratification, and it predicts future cirrhosis and all-cause mortality in MASLD (176,197,205,206,218). An LSM value of <8.0 kPa has a good negative predictive value to exclude advanced fibrosis ($\geq F3$ –F4) (219–221) and indicates lower risk for clinically significant fibrosis. Such individuals with prediabetes or type 2 diabetes can be followed in nonspecialty clinics with repeat surveillance testing every ≥ 2 years, although the precise time interval remains to be established. If the LSM is ≥ 8.0 kPa, the risk for advanced fibrosis ($\geq F3$ –F4) is higher and such individuals should be referred to the hepatologist (176,182,197,205) within the framework of an interprofessional team (222–224). FIB-4 followed by LSM helps stratify individuals with diabetes by risk level and minimize specialty referrals (218,225–228) (Fig. 4.2). Given the lack of widespread availability of LSM, the ELF test is a good alternative (229). Individuals with ELF <9.8 are considered at low risk for adverse liver outcomes. Individuals with ELF ≥ 9.8 are considered at high risk of having MASH with advanced liver fibrosis ($\geq F3$ –F4) and therefore are at risk for adverse liver outcomes (176,206,208).

Diagnostic algorithm for the prevention of cirrhosis in people with metabolic dysfunction–associated steatotic liver disease (MASLD)

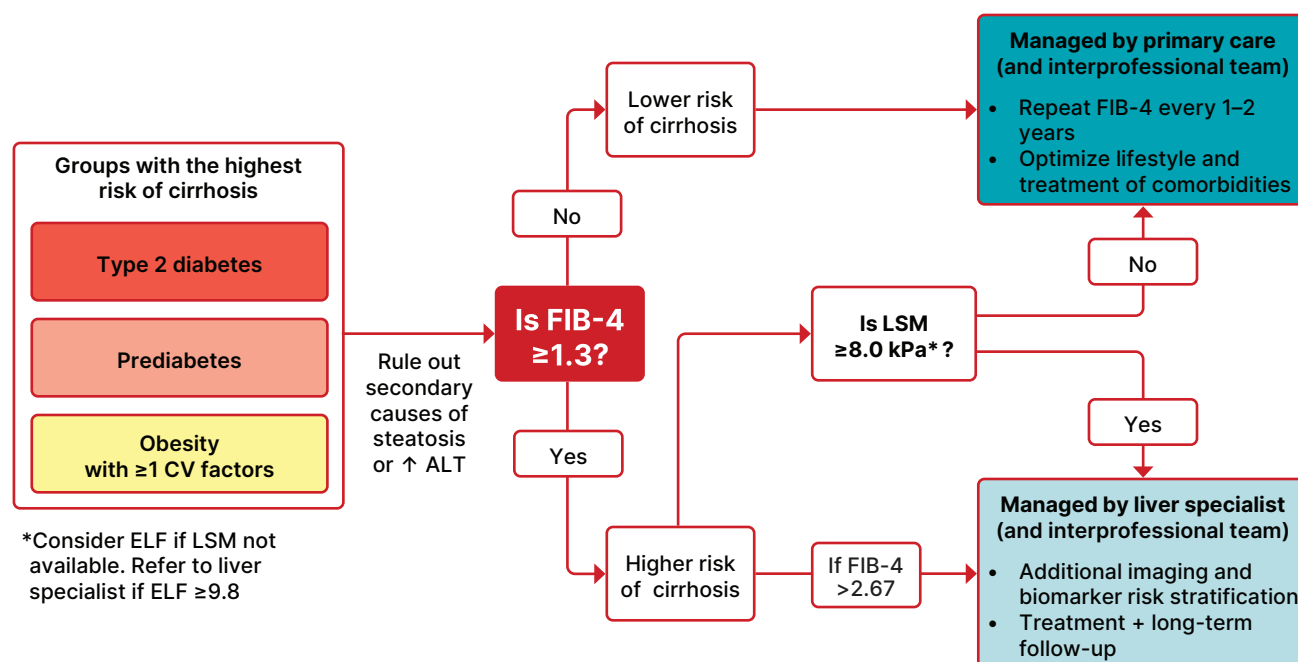


Figure 4.2—Diagnostic algorithm for risk stratification and the prevention of cirrhosis in individuals with metabolic dysfunction-associated steatotic liver disease (MASLD). CV, cardiovascular; ELF, enhanced liver fibrosis test; FIB-4, fibrosis-4 index; LSM, liver stiffness measurement, as measured by vibration-controlled transient elastography. *In the absence of LSM, consider ELF a diagnostic alternative. If ELF ≥ 9.8 , an individual is at high risk of metabolic dysfunction-associated steatohepatitis with advanced liver fibrosis ($\geq F3$ –F4) and should be referred to a liver specialist.

They should be referred to a gastroenterologist or hepatologist. The optimal cutoff for clinical use of ELF in primary care and endocrinology settings is evolving (230–233). An ELF <9.8 suggests an individual is at low risk of advanced liver fibrosis and may be followed in the nonspecialty clinic with repeat testing in ≥ 2 years but may need repeat testing more often if ELF is between 9.2 and 9.7.

Specialists may order additional tests for fibrosis risk stratification in MASH (175,197,205,208), including magnetic resonance elastography (MRE) (best overall performance, particularly for early fibrosis stages) or multiparametric iron-corrected T1 MRI (cT1) (234) and patented blood-based fibrosis biomarkers. While liver biopsy remains the gold standard for the diagnosis of MASH, its indication is reserved to the discretion of the specialist within an interprofessional team approach due to high costs and potential for morbidity associated with this procedure.

Management

Recommendations

4.25 Adults with type 2 diabetes or prediabetes, particularly with overweight or obesity, who have metabolic dysfunction–associated steatotic liver disease (MASLD) should be recommended lifestyle changes using an interprofessional approach that promotes weight loss, ideally within a structured nutrition plan and physical activity program for cardiometabolic benefits **B** and histological improvement. **C**

4.26 In adults with type 2 diabetes, MASLD, and overweight or obesity, consider using a glucagon-like peptide 1 receptor agonist (GLP-1 RA) with demonstrated benefits in MASH **A** or a dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RA with potential benefits in MASH **B** for the treatment of obesity as an adjunctive therapy to lifestyle interventions for weight loss.

4.27a In adults with type 2 diabetes and biopsy-proven MASH or those at high risk for liver fibrosis (based on noninvasive tests), a GLP-1 RA is preferred for glycemic management due to beneficial effects on MASH. **A** Pioglitazone **B** or a dual GIP and GLP-1 RA **B** can be considered for

glycemic management due to potential beneficial effects on MASH.

4.27b Combination therapy with pioglitazone plus GLP-1 RA can be considered for the treatment of hyperglycemia in adults with type 2 diabetes with biopsy-proven MASH or those at high risk of liver fibrosis (identified with noninvasive tests) because of potential beneficial effects on MASH. **B**

4.28 For consideration of treatment with a thyroid hormone receptor- β agonist in adults with type 2 diabetes or prediabetes with MASLD with moderate (F2) or advanced (F3) liver fibrosis on liver histology, or by a validated imaging-based or blood-based test, refer to a gastroenterologist or hepatologist with expertise in MASLD management. **A**

4.29 Treatment initiation and monitoring should be individualized and within the context of an interprofessional team that includes a gastroenterologist or hepatologist, consideration of individual preferences, and a careful shared-decision cost-benefit discussion. **B**

4.30a In adults with type 2 diabetes and MASLD, use of glucose-lowering therapies other than pioglitazone or GLP-1 RAs may be continued as clinically indicated, but these therapies lack evidence of benefit in MASH. **B**

4.30b Insulin therapy is the preferred agent for the treatment of hyperglycemia in adults with type 2 diabetes with decompensated cirrhosis. **C**

4.31a Adults with type 2 diabetes and MASLD are at increased cardiovascular risk; therefore, comprehensive management of cardiovascular risk factors is recommended. **B**

4.31b Statin therapy is safe in adults with type 2 diabetes and compensated cirrhosis from MASLD and should be initiated or continued for cardiovascular risk reduction as clinically indicated.

B In people with decompensated cirrhosis, statin therapy should be used with caution, and close monitoring is needed, given limited safety and efficacy data. **B**

4.32a Consider metabolic surgery in appropriate candidates as an option to treat MASH in adults with type 2

diabetes and obesity **B** and to improve cardiovascular outcomes. **B**

4.32b Metabolic surgery should be used with caution in adults with type 2 diabetes with compensated cirrhosis from MASLD **B** and is not recommended in decompensated cirrhosis. **B**

While steatohepatitis and cirrhosis occur in lean people with diabetes and are believed to be linked to genetic predisposition, insulin resistance, and environmental factors (235,236), ample evidence implicates excess visceral fat and overall adiposity in individuals with overweight and obesity in the pathogenesis of the disease (236,237). Obesity in the setting of type 2 diabetes worsens insulin resistance and steatohepatitis, promoting the development of cirrhosis (238). Therefore, clinicians should enact evidence-based interventions (as discussed in section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes”) to promote healthy lifestyle change and weight loss for individuals with overweight or obesity and MASLD. There is consensus that a minimum weight loss goal of 5%, preferably $\geq 10\%$, is needed to improve liver histology (176,197,205,208), with fibrosis requiring the larger weight reduction to promote change (239,240). However, there is significant individual variability in histological outcomes with weight loss. Individualized, structured weight loss and exercise programs offer greater benefit than standard counseling in people with MASLD (241).

Nutrition plan recommendations to induce an energy deficit are not different from those for individuals with diabetes with obesity without MASLD and should include a reduction of macronutrient content, limiting saturated fat, refined carbohydrates, and added sugar, with adoption of healthier eating patterns. The Mediterranean eating pattern has the best evidence for improving liver and cardiometabolic health (205–208,241,242). Both aerobic and resistance training improve MASLD in proportion to treatment engagement and intensity of the program (243). Obesity pharmacotherapy may assist with weight loss in the context of lifestyle modification if not achieved by lifestyle modification alone (see section 8,

“Obesity and Weight Management for the Prevention and Treatment of Diabetes”).

Given the high prevalence of at-risk MASH (~12–20%) (178,179,181,182), higher risk of disease progression and liver-related mortality (180,196,244), and the lack of pharmacological treatments once cirrhosis is established (245,246), optimizing the pharmacological management of hyperglycemia and obesity in individuals with type 2 diabetes and MASH could serve the dual purpose of addressing these comorbidities while treating the liver disease (Fig. 4.3). Therefore, early diagnosis and treatment of MASLD offers the best opportunity for cirrhosis prevention. In phase 2 clinical trials, pioglitazone and some GLP-1 RAs have been shown to be potentially effective to treat steatohepatitis (205,247–250) and to slow fibrosis progression (251,252). They may also decrease CVD (248), which is the number one cause of death in individuals with type 2 diabetes and MASLD (200).

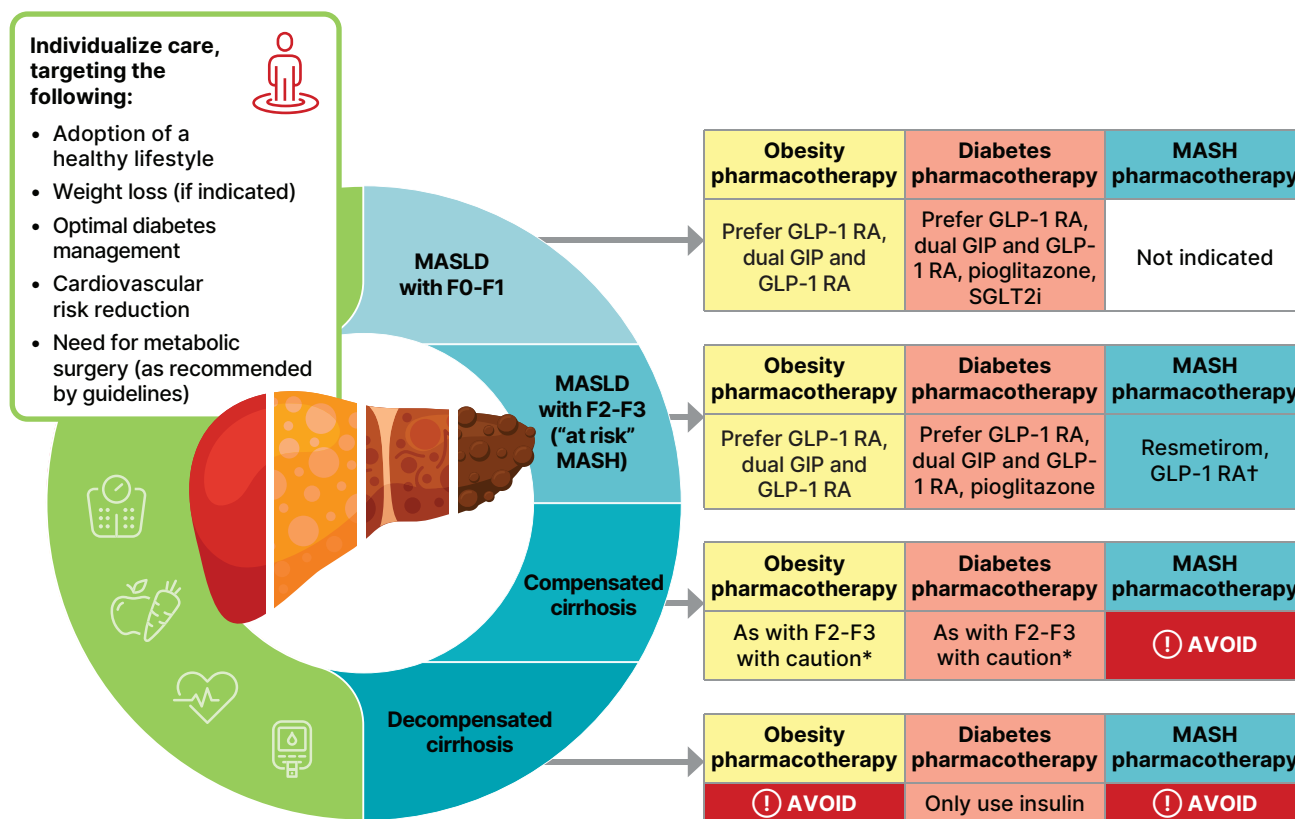
Recently, a phase 3 clinical trial examined the effect of semaglutide in subjects with MASH without cirrhosis (ESSENCE) (253). The trial reported that in 36.8% of individuals treated for 72 weeks with semaglutide 2.4 mg subcutaneously once weekly, improvement in liver fibrosis was achieved with no worsening of steatohepatitis, compared with 22.4% on placebo ($P < 0.001$), while 62.9% achieved resolution of steatohepatitis with no worsening of liver fibrosis, compared with 34.3% on placebo ($P < 0.001$). Based on the phase 3 trial results, the FDA recently approved semaglutide for the treatment of MASH with moderate to advanced liver fibrosis (254).

Semaglutide is currently the only glucose-lowering or obesity medication that is FDA approved for the treatment of MASH. The recommendation to treat hyperglycemia with other GLP-1 RA–based therapies and/or pioglitazone in individuals with type 2 diabetes and MASLD is

based on consistent histological benefit for steatohepatitis in several phase 2 randomized controlled trials (RCTs) with GLP-1 RAs (254–259) and with pioglitazone (255,256) compared with no benefit with metformin or other glucose-lowering medications in MASH (176,197,205).

Pioglitazone improves glucose and lipid metabolism and reverses steatohepatitis in individuals with prediabetes or type 2 diabetes (252,255,257) and even in individuals without diabetes (256,258,259) (Fig. 4.3). Fibrosis also improved in some trials (247,259). A meta-analysis (251) concluded that pioglitazone treatment results in resolution of MASH and may improve fibrosis. Furthermore, combination therapy with pioglitazone plus a GLP-1 RA has been reported safe and effective for the treatment of hyperglycemia in adults with type 2 diabetes (260–263) as well as in reducing hepatic steatosis (260,262), suggesting additive benefit in individuals with MASLD. It is important

Metabolic dysfunction-associated steatotic liver disease (MASLD) treatment algorithm



† Only semaglutide among GLP-1 RAs has been approved by the FDA for treatment of MASH.

* Individualized care and close monitoring needed in compensated cirrhosis given limited safety data available.

Figure 4.3—Metabolic dysfunction-associated steatotic liver disease (MASLD) treatment algorithm. F0-F1, no to minimal fibrosis; F2-F3, moderate fibrosis; F4, cirrhosis; GIP, glucose-dependent insulintropic polypeptide; GLP-1 RA, glucagon-like peptide 1 receptor agonist; MASH, metabolic dysfunction-associated steatohepatitis; SGLT2i, sodium–glucose cotransporter 2 inhibitor.

to note that these studies are based on phase 2 clinical trials and await further phase 3 evidence. However, these plans are attractive because they offer potential benefit compared with lack of histological benefit (or clinical trial data) from other oral glucose-lowering therapies in MASLD. In the context of treating hyperglycemia in individuals with type 2 diabetes with MASLD, where the low cost of pioglitazone and any liver improvement would be an added benefit to glycemic management, these plans would be potentially cost-effective for the treatment of MASLD (264,265). Vitamin E may be beneficial for the treatment of MASH in individuals without diabetes (258). However, in individuals with type 2 diabetes, vitamin E monotherapy was found to be ineffective in a small RCT (252), and it did not seem to enhance pioglitazone's efficacy when used in combination, as reported in an earlier trial in this population (257). Pioglitazone causes dose-dependent weight gain (15 mg/day, mean weight gain of 1–2%; 45 mg/day, mean weight gain of 3–5%), which can be blunted or reversed if combined with SGLT2 inhibitors or GLP-1 RAs (248,262,263,266). Pioglitazone increases fracture risk, may promote heart failure if used in individuals with preexisting congestive heart failure, and may increase the risk of bladder cancer, although this remains controversial (176,197,205,248,249).

GLP-1 RAs are effective at inducing weight loss and ameliorating elevated plasma aminotransferases and steatosis (247) (**Fig. 4.3**). As discussed above, semaglutide was effective as a treatment for MASH with moderate to advanced liver fibrosis in a phase 3 RCT (253). However, there are few phase 2 RCTs of other GLP-1 RAs in individuals with MASH proven by biopsy. A small RCT reported that liraglutide improved some features of MASH and may delay fibrosis progression (267). Tirzepatide is a dual GIP and GLP-1 RA known to reduce liver steatosis in MASLD (268), and a phase 2 paired-biopsy study of 190 adults with overweight or obesity with MASH (50–60% of whom had type 2 diabetes) recently reported that doses of 5, 10, and 15 mg/day resulted in resolution of steatohepatitis without worsening of fibrosis in 44%, 56%, and 62% of participants, respectively, compared with 10% of participants receiving placebo ($P < 0.001$ for all three comparisons) (269). Improvement of at least one fibrosis stage

without worsening of MASH occurred in 55%, 51%, and 51% of participants, respectively, compared with 30% of participants receiving placebo. Survodutide is a dual GLP-1 and glucagon receptor agonist that is in development, and a phase 2 paired-biopsy trial recently reported benefit in MASH (270). In summary, other GLP-1-based therapies (besides semaglutide) and/or pioglitazone are recommended to treat type 2 diabetes in adults with MASH based on histological benefit for steatohepatitis in several phase 2 RCTs (269,270) compared with no benefit with metformin or other glucose-lowering or obesity medications. Within the context of their approved indication (e.g., obesity or type 2 diabetes), these medications are cost-effective to treat the comorbidity, while potentially improving MASH, which becomes an added benefit.

SGLT2 inhibitors (271–273) and insulin (249) reduce hepatic steatosis, but their effects on steatohepatitis remain unknown. The use of glucose-lowering agents other than pioglitazone or GLP-1 RAs may be continued in individuals with type 2 diabetes and MASLD for glycemic management, as clinically indicated. However, these agents have either failed to improve steatohepatitis in paired-biopsy studies (metformin) or have no RCTs with liver histological end points (i.e., sulfonylureas, glitinides, dipeptidyl peptidase 4 inhibitors, or acarbose).

Resmetirom is a thyroid hormone receptor- β agonist approved by the FDA for the treatment of adults with MASLD with moderate (F2) or advanced (F3) liver fibrosis on liver histology or a validated imaging- or blood-based test. In a phase 3 RCT, resmetirom for 52 weeks in 966 adults at the highest dose of 100 mg (or placebo) met the primary end point of MASH resolution without worsening of fibrosis in 29.9% of individuals compared with 9.7% on placebo ($P < 0.001$) (274). Fibrosis improved in up to 25.9% and 14.2%, respectively ($P < 0.001$). Nausea, vomiting, and diarrhea occurred more often with resmetirom. The gastrointestinal side effects are dose dependent and improve with continued treatment. Resmetirom decreased free thyroxine (T4) levels by ~20% and increased sex hormone-binding protein levels two- to threefold. Although a recent review of the data concluded that there is little concern about these changes, long-term postmarketing data must be collected (275). Guidance by the

American Association for the Study of Liver Diseases (AASLD) about optimal individual identification for treatment, safety, and long-term monitoring has recently been published (276). This is especially relevant because hypothyroidism and hypogonadism are more prevalent in people with MASLD than in the general population (176,205), and clinicians should monitor all individuals with MASLD for symptoms of endocrine deficiency and manage according to clinical practice guidelines. Per its label, candidates for resmetirom treatment are those with MASLD and moderate (F2) to advanced (F3) liver fibrosis but not with cirrhosis or other active liver disease (i.e., alcohol-related liver disease, autoimmune hepatitis, or primary biliary cholangitis) or inadequately managed hypothyroidism or hyperthyroidism. Given complexities associated with selection of an individual for therapy, drug cost, and treatment monitoring, therapy should be individualized and initiated by a hepatologist or gastroenterologist with expertise in MASH within an interprofessional team.

Insulin is the preferred glucose-lowering agent for the treatment of hyperglycemia in adults with type 2 diabetes with decompensated cirrhosis given the lack of robust evidence about the safety and efficacy of oral agents and noninsulin injectables (i.e., GLP-1 RAs and dual GIP and GLP-1 RAs) (246), although a recent 48-week study suggested that GLP-1 RAs are safe in individuals with MASH and compensated cirrhosis (277).

Metabolic surgery leading to sustained weight loss and improvement of type 2 diabetes can improve MASH and cardiometabolic health, altering the natural history of the disease (278). Meta-analyses report that 70–80% of people have improvement in hepatic steatosis, 50–75% of people have improvement in inflammation and hepatocyte ballooning (necrosis), and 30–40% of people have improvement in fibrosis (279,280). It may also reduce the risk of HCC (280). It is important to note that currently metabolic surgery is not indicated solely for treatment of MASH. Given that many individuals with MASH have metabolic risks (type 2 diabetes and obesity) that are indications for metabolic surgery, the improvement in liver health is expected, but surgical indication should follow current practice guidelines. Metabolic surgery should be used with caution in individuals with compensated cirrhosis (i.e., asymptomatic stage of

cirrhosis without associated liver complications), but with experienced surgeons the risk of hepatic decompensation is similar to that for individuals with less advanced liver disease. Because of the paucity of safety and outcome data, metabolic surgery is not recommended in individuals with decompensated cirrhosis (i.e., cirrhosis stage with complications such as variceal hemorrhage, ascites, hepatic encephalopathy, or jaundice) who also have a much higher risk of postoperative development of these liver-related complications (176,197,205).

Adults with type 2 diabetes and MASLD are at an increased risk of CVD and require comprehensive management of cardiovascular risk factors (176,197,205). Within an interprofessional approach, statin therapy should be initiated or continued for cardiovascular risk reduction as clinically indicated. Overall, its use appears to be safe in adults with type 2 diabetes and MASH, including in the presence of compensated cirrhosis (Child-Pugh class A or B cirrhosis) from MASLD. Some studies even suggest that statin use in individuals with chronic liver disease may reduce episodes of hepatic decompensation and/or overall mortality (281,282). Statin therapy is not recommended in decompensated cirrhosis given limited safety and efficacy data (176,197,206).

Obstructive Sleep Apnea

Age-adjusted rates of obstructive sleep apnea, a risk factor for CVD, are significantly higher (4- to 10-fold) with obesity, especially with central obesity (283) (see section 5, "Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes"). The prevalence of obstructive sleep apnea in the population with type 2 diabetes may be as high as 23%, and the prevalence of any sleep-disordered breathing may be as high as 58% (284,285). In participants with obesity and type 2 diabetes enrolled in the Look AHEAD trial, the prevalence exceeded 80% (286). Obstructive sleep apnea should be evaluated in individuals with suggestive symptoms (e.g., excessive daytime sleepiness, snoring, and witnessed apnea) (287). Sleep apnea treatment (lifestyle modification, continuous positive airway pressure, oral appliances, and surgery) significantly improves quality of life and blood pressure management. Recently, two phase 3 randomized trials found that among adults

with obesity and moderate-to-severe obstructive sleep apnea but without diabetes, treatment with the dual GIP and GLP-1 RA tirzepatide substantially reduced sleep apnea severity (288). More research is needed to determine the effects of GLP-1 and dual GIP and GLP-1 RAs on sleep apnea in individuals with diabetes.

Pancreatitis

Diabetes is linked to diseases of the exocrine pancreas, such as pancreatitis, which may disrupt the global architecture or physiology of the pancreas, often resulting in both exocrine and endocrine dysfunction. Up to half of individuals with diabetes may have some degree of impaired exocrine pancreas function (289). People with diabetes are at an approximately twofold higher risk of developing acute pancreatitis (290).

Conversely, prediabetes and/or diabetes has been found to develop in approximately one-third of individuals after an episode of acute pancreatitis (291); thus, the relationship is likely bidirectional. Post-pancreatitis diabetes may include either new-onset disease or previously unrecognized diabetes (292). Studies of individuals treated with incretin-based therapies for diabetes have also reported that pancreatitis may occur more frequently with these medications, but results have been mixed and causality has not been established (293,294).

Islet autotransplantation should be considered for individuals requiring total pancreatectomy for medically refractory chronic pancreatitis to prevent postsurgical diabetes. Approximately one-third of individuals undergoing total pancreatectomy with islet autotransplantation are insulin free 1 year postoperatively, and observational studies from different centers have demonstrated islet graft function up to a decade after the surgery in some individuals (294–298). Both personal factors for the individual with diabetes and disease factors should be carefully considered when deciding the indications and timing of this surgery. Surgeries should be performed in skilled facilities that have demonstrated expertise in islet autotransplantation.

Sensory Impairment

Hearing impairment, both in high-frequency and low- to midfrequency ranges, is more common in people with diabetes than in

those without, with stronger associations found in studies of younger people (299). Proposed pathophysiologic mechanisms include the combined contributions of hyperglycemia and oxidative stress with cochlear microangiopathy and auditory neuropathy (300). In a National Health and Nutrition Examination Survey (NHANES) analysis, hearing impairment was about twice as prevalent in individuals with diabetes as in those without, after adjusting for age and other risk factors for hearing impairment (301). Low HDL cholesterol, coronary heart disease, peripheral neuropathy, and general poor health have been reported as risk factors for hearing impairment for individuals with diabetes, but an association of hearing loss with glycemia has not been consistently observed (302). In the DCCT/EDIC cohort, increases in the time-weighted mean A1C was associated with increased risk of hearing impairment when tested after long-term (>20 years) follow-up, with every 10% increase in A1C leading to 19% high-frequency impairment (303). Impairment in smell, but not taste, has also been reported in individuals with diabetes (304).

References

- Goh LH, Siah CJR, Tam WWS, Tai ES, Young DY. Effectiveness of the chronic care model for adults with type 2 diabetes in primary care: a systematic review and meta-analysis. *Syst Rev* 2022;11:273
- Rodriguez K, Ryan D, Dickinson JK, Phan V. Improving quality outcomes: the value of diabetes care and education specialists. *Clin Diabetes* 2022; 40:356–365
- Dickinson JK, Guzman SJ, Maryniuk MD, et al. The use of language in diabetes care and education. *Diabetes Care* 2017;40:1790–1799
- Ji M, Sereika SM, Dunbar-Jacob J, Erlen JA. Correlation of symptom distress, self-efficacy, and social support with problem-solving and glycemic control among patients with type 2 diabetes. *Sci Diabetes Self Manag Care* 2021;47: 85–93
- Beverly EA, Hughes AS, Saunders A. Examination of health care providers' use of language in diabetes care: a secondary qualitative data analysis. *Clin Diabetes* 2022;40:434–441
- Issa AN, Wodi AP, Moser CA, Cineas S. Advisory committee on immunization practices recommended immunization schedule for children and adolescents aged 18 years or younger - United States, 2025. *MMWR Morb Mortal Wkly Rep* 2025;74:26–29
- Wodi AP, Issa AN, Moser CA, Cineas S. Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older - United States, 2025. *MMWR Morb Mortal Wkly Rep* 2025;74:30–33
- Centers for Disease Control and Prevention, Advisory Committee on Immunization Practice (ACIP). ACIP Evidence to Recommendation User's

- Guide, 2020. Accessed 25 August 2025. Available from <https://stacks.cdc.gov/view/cdc/127248>
9. Goeijenbier M, van Sloten TT, Slobbe L, et al. Benefits of flu vaccination for persons with diabetes mellitus: a review. *Vaccine* 2017;35:5095–5101
10. Behrouzi B, Bhatt DL, Cannon CP, et al. Association of influenza vaccination with cardiovascular risk: a meta-analysis. *JAMA Netw Open* 2022;5:e228873
11. Kornum JB, Thomsen RW, Riis A, Lervang H-H, Schønheyder HC, Sørensen HT. Diabetes, glycemic control, and risk of hospitalization with pneumonia: a population-based case-control study. *Diabetes Care* 2008;31:1541–1545
12. Centers for Disease Control and Prevention. Expanded Recommendations for Use of Pneumococcal Conjugate Vaccines Among Adults Aged ≥50 Years: recommendations of the Advisory Committee on Immunization Practices — United States, 2024. *Morbidity and Mortality Weekly Report (MMWR)* 2025;74:1–8
13. Centers for Disease Control and Prevention. Advisory Committee on Immunization Practices Recommended Immunization Schedule for Adults Aged 19 Years or Older — United States, 2025. *MMWR Morb Mortal Wkly Rep* 2025;74:30–33
14. Britton A, Roper LE, Kotton CN, et al. Use of respiratory syncytial virus vaccines in adults aged ≥60 years: updated recommendations of the Advisory Committee on Immunization Practices - United States, 2024. *MMWR Morb Mortal Wkly Rep* 2024;73:696–702
15. Tinetti ME, Fried TR, Boyd CM. Designing health care for the most common chronic condition—multimorbidity. *JAMA* 2012;307:2493–2494
16. Sudore RL, Karter AJ, Huang ES, et al. Symptom burden of adults with type 2 diabetes across the disease course: diabetes & aging study. *J Gen Intern Med* 2012;27:1674–1681
17. Nederstigt C, Uitbeijerse BS, Janssen LGM, Corssmit EPM, de Koning EJP, Dekkers OM. Associated auto-immune disease in type 1 diabetes patients: a systematic review and meta-analysis. *Eur J Endocrinol* 2019;180:135–144
18. Hughes JW, Riddlesworth TD, DiMeglio LA, Miller KM, Rickels MR, McGill JB; T1D Exchange Clinic Network. Autoimmune diseases in children and adults with type 1 diabetes from the T1D Exchange Clinic Registry. *J Clin Endocrinol Metab* 2016;101:4931–4937
19. Kahaly GJ, Hansen MP. Type 1 diabetes associated autoimmunity. *Autoimmun Rev* 2016;15:644–648
20. Eisenbarth GS, Gottlieb PA. Autoimmune polyendocrine syndromes. *N Engl J Med* 2004;350:2068–2079
21. Pham-Short A, Donaghue KC, Ambler G, Phelan H, Twigg S, Craig ME. Screening for celiac disease in type 1 diabetes: a systematic review. *Pediatrics* 2015;136:e170–e176
22. Cauley JA, Hochberg MC, Lui L-Y, et al. Long-term risk of incident vertebral fractures. *JAMA* 2007;298:2761–2767
23. Kanis JA, Johnell O, De Laet C, et al. A meta-analysis of previous fracture and subsequent fracture risk. *Bone* 2004;35:375–382
24. Pedersen AB, Ehrenstein V, Szépligeti SK, et al. Thirty-five-year trends in first-time hospitalization for hip fracture, 1-year mortality, and the prognostic impact of comorbidity: a Danish Nationwide Cohort Study, 1980–2014. *Epidemiology* 2017;28:898–905
25. Tajeu GS, Delzell E, Smith W, et al. Death, disability, and destitution following hip fracture. *J Gerontol A Biol Sci Med Sci* 2014;69:346–353
26. Miao J, Brismar K, Nyrén O, Ugargh-Morawski A, Ye W. Elevated hip fracture risk in type 1 diabetic patients: a population-based cohort study in Sweden. *Diabetes Care* 2005;28:2850–2855
27. Wang H, Ba Y, Xing Q, Du J-L. Diabetes mellitus and the risk of fractures at specific sites: a meta-analysis. *BMJ Open* 2019;9:e024067
28. Weber DR, Haynes K, Leonard MB, Willi SM, Denburg MR. Type 1 diabetes is associated with an increased risk of fracture across the life span: a population-based cohort study using The Health Improvement Network (THIN). *Diabetes Care* 2015;38:1913–1920
29. Leanza G, Maddaloni E, Pitocco D, et al. Risk factors for fragility fractures in type 1 diabetes. *Bone* 2019;125:194–199
30. Janghorbani M, Van Dam RM, Willett WC, Hu FB. Systematic review of type 1 and type 2 diabetes mellitus and risk of fracture. *Am J Epidemiol* 2007;166:495–505
31. Napoli N, Schwartz AV, Schafer AL, et al.; Osteoporotic Fractures in Men (MrOS) Study Research Group. Vertebral fracture risk in diabetic elderly men: the MrOS study. *J Bone Miner Res* 2018;33:63–69
32. Koromani F, Oei L, Shevroja E, et al. Vertebral fractures in individuals with type 2 diabetes: more than skeletal complications alone. *Diabetes Care* 2020;43:137–144
33. Faraj M, Schwartz AV, Burghardt AJ, et al. Risk factors for bone microarchitecture impairments in older men with type 2 diabetes - the MrOS study. *J Clin Endocrinol Metab* 2025;110:e1660–e1669
34. Schwartz AV, Vittinghoff E, Bauer DC, et al.; Study of Osteoporotic Fractures (SOF) Research Group; Osteoporotic Fractures in Men (MrOS) Research Group; Health, Aging, and Body Composition (Health ABC) Research Group. Association of BMD and FRAX score with risk of fracture in older adults with type 2 diabetes. *JAMA* 2011;305:2184–2192
35. Hidayat K, Fang Q-L, Shi B-M, Qin L-Q. Influence of glycemic control and hypoglycemia on the risk of fracture in patients with diabetes mellitus: a systematic review and meta-analysis of observational studies. *Osteoporos Int* 2021;32:1693–1704
36. Wang B, Wang Z, Poundarik AA, et al. Unmasking fracture risk in type 2 diabetes: the association of longitudinal glycemic hemoglobin level and medications. *J Clin Endocrinol Metab* 2022;107:e1390–e1401
37. Komorita Y, Iwase M, Fujii H, et al. Both hypo- and hyperglycaemia are associated with increased fracture risk in Japanese people with type 2 diabetes: the Fukuoka Diabetes Registry. *Diabet Med* 2020;37:838–847
38. Majumdar SR, Leslie WD, Lix LM, et al. Longer duration of diabetes strongly impacts fracture risk assessment: the Manitoba BMD cohort. *J Clin Endocrinol Metab* 2016;101:4489–4496
39. Van Hulten V, Driessen JHM, Andersen S, et al. Fracture risk revisited: bone mineral density T-score and fracture risk in type 2 diabetes. *Diabetes Obes Metab* 2024;26:5325–5335
40. Napoli N, Strollo R, Paladini A, Briganti SI, Pozzilli P, Epstein S. Corrigendum to “The Alliance of Mesenchymal Stem Cells, Bone, and Diabetes. *Int J Endocrinol* 2017;2017:5924671
41. Vavanikunnel J, Charlier S, Becker C, et al. Association between glycemic control and risk of fracture in diabetic patients: a nested case-control study. *J Clin Endocrinol Metab* 2019;104:1645–1654
42. Strotmeyer ES, Cauley JA, Schwartz AV, et al. Nontraumatic fracture risk with diabetes mellitus and impaired fasting glucose in older white and black adults: the health, aging, and body composition study. *Arch Intern Med* 2005;165:1612–1617
43. Schwartz AV, Vittinghoff E, Sellmeyer DE, et al.; Health, Aging, and Body Composition Study. Diabetes-related complications, glycemic control, and falls in older adults. *Diabetes Care* 2008;31:391–396
44. Piccoli A, Cannata F, Strollo R, et al. Sclerostin regulation, microarchitecture, and advanced glycation end-products in the bone of elderly women with type 2 diabetes. *J Bone Miner Res* 2020;35:2415–2422
45. Leanza G, Cannata F, Faraj M, et al. Bone canonical Wnt signaling is downregulated in type 2 diabetes and associates with higher advanced glycation end-products (AGEs) content and reduced bone strength. *Elife* 2024;12:RP90437
46. Tramontana F, Napoli N, Litwack-Harrison S, et al. More rapid bone mineral density loss in older men with diabetes: the Osteoporotic Fractures in Men (MrOS) Study. *J Clin Endocrinol Metab* 2024;109:e2283–e2290
47. Ferrari SL, Abrahamsen B, Napoli N, et al.; Bone and Diabetes Working Group of IOF. Diagnosis and management of bone fragility in diabetes: an emerging challenge. *Osteoporos Int* 2018;29:2585–2596
48. Leslie WD, Johansson H, McCloskey EV, Harvey NC, Kanis JA, Hans D. Comparison of methods for improving fracture risk assessment in diabetes: the Manitoba BMD Registry. *J Bone Miner Res* 2018;33:1923–1930
49. Ferrari S, Eastell R, Napoli N, et al. Denosumab in postmenopausal women with osteoporosis and diabetes: subgroup analysis of FREEDOM and FREEDOM extension. *Bone* 2020;134:115268
50. Napoli N, Conte C, Eastell R, et al. Bone turnover markers do not predict fracture risk in type 2 diabetes. *J Bone Miner Res* 2020;35:2363–2371
51. International Society for Pediatric and Adolescent Diabetes. ISPAD Clinical Practice Consensus Guidelines 2022. Accessed 25 August 2025. Available from <https://www.ispad.org/resources/ispad-clinical-practice-consensus-guidelines/2022-ispad-clinical-practice-consensus-guidelines/ispad-cpcg-2022-english.html>
52. LeBoff MS, Greenspan SL, Insogna KL, et al. The clinician's guide to prevention and treatment of osteoporosis. *Osteoporos Int* 2022;33:2049–2102
53. Armamento-Villareal R, Aguirre L, Napoli N, et al. Changes in thigh muscle volume predict bone mineral density response to lifestyle therapy in frail, obese older adults. *Osteoporos Int* 2014;25:551–558
54. Ebeling PR, Adler RA, Jones G, et al. Management of endocrine disease: therapeutics of vitamin D. *Eur J Endocrinol* 2018;179:R239–R259
55. Maddaloni E, Cavallari I, Napoli N, Conte C. Vitamin D and diabetes mellitus. *Front Horm Res* 2018;50:161–176
56. Iolascon G, Gimigliano R, Bianco M, et al. Are dietary supplements and nutraceuticals effective for musculoskeletal health and cognitive function?

- A scoping review. *J Nutr Health Aging* 2017;21:527–538
57. National Institutes of Health. Calcium—fact sheet for health professionals. Accessed 27 August 2025. Available from <https://ods.od.nih.gov/factsheets/Calcium-HealthProfessional>
 58. Rosen CJ, Abrams SA, Aloia JF, et al. IOM committee members respond to Endocrine Society vitamin D guideline. *J Clin Endocrinol Metab* 2012;97:1146–1152
 59. National Institutes of Health. Vitamin D—fact sheet for health professionals. Accessed 27 August 2025. Available from <https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional>
 60. Carmel AS, Shieh A, Bang H, Bockman RS. The 25(OH)D level needed to maintain a favorable bisphosphonate response is ≥ 33 ng/ml. *Osteoporos Int* 2012;23:2479–2487
 61. Holick MF, Siris ES, Binkley N, et al. Prevalence of vitamin D inadequacy among postmenopausal North American women receiving osteoporosis therapy. *J Clin Endocrinol Metab* 2005;90:3215–3224
 62. U.S. Preventive Services Task Force. Vitamin D, Calcium, or Combined Supplementation for the Primary Prevention of Falls and Fractures in Community-Dwelling Adults: preventive Medication. Accessed 25 September 2025. Available from <https://www.uspreventiveservicestaskforce.org/uspstf/draft-recommendation/vitamin-d-calcium-combined-supplementation-primary-prevention-falls-fractures-communitydwelling-adults>
 63. Sinclair AJ, Abdelhafiz A, Dunning T, et al. An international position statement on the management of frailty in diabetes mellitus: summary of recommendations 2017. *J Frailty Aging* 2018;7:10–20
 64. Eastell R, Vittinghoff E, Lui L-Y, et al. Diabetes mellitus and the benefit of antiresorptive therapy on fracture risk. *J Bone Miner Res* 2022;37:2121–2131
 65. Langdahl BL, Silverman S, Fujiwara S, et al. Real-world effectiveness of teriparatide on fracture reduction in patients with osteoporosis and comorbidities or risk factors for fractures: integrated analysis of 4 prospective observational studies. *Bone* 2018;116:58–66
 66. Schwartz AV, Pavo I, Alam J, et al. Teriparatide in patients with osteoporosis and type 2 diabetes. *Bone* 2016;91:152–158
 67. Napoli N, Pannaciuoli N, Vittinghoff E, et al. Effect of denosumab on fasting glucose in women with diabetes or prediabetes from the FREEDOM trial. *Diabetes Metab Res Rev* 2018;34:e2991
 68. Tsur A, Cahn A, Levy L, Pollack R. Cardiovascular outcomes of romosozumab treatment-real-world data analysis. *JBMR Plus* 2025;9:z1ae146
 69. Langdahl BL, Hofbauer LC, Forfar JC. Cardiovascular safety and sclerostin inhibition. *J Clin Endocrinol Metab* 2021;106:1845–1853
 70. Cosman F, Crittenden DB, Adachi JD, et al. Romosozumab treatment in postmenopausal women with osteoporosis. *N Engl J Med* 2016;375:1532–1543
 71. Conley RB, Adib G, Adler RA, et al. Secondary fracture prevention: consensus clinical recommendations from a multistakeholder coalition. *J Bone Miner Res* 2020;35:36–52
 72. Hofbauer LC, Rachner TD. More DATA to guide sequential osteoporosis therapy. *Lancet* 2015;386:1116–1118
 73. Palermo A, D'Onofrio L, Eastell R, Schwartz AV, Pozzilli P, Napoli N. Oral anti-diabetic drugs and fracture risk, cut to the bone: safe or dangerous? A narrative review. *Osteoporos Int* 2015;26:2073–2089
 74. Maddaloni E, Coleman RL, Holman RR. Risk factors for bone fractures in type 2 diabetes and the impact of once-weekly exenatide: insights from an EXSCEL post-hoc analysis. *Diabetes Res Clin Pract* 2025;223:112125
 75. Loke YK, Singh S, Furburg CD. Long-term use of thiazolidinediones and fractures in type 2 diabetes: a meta-analysis. *CMAJ* 2009;180:32–39
 76. Dormandy J, Bhattacharya M, van Troostenburg de Bruyn A-R, PROactive Investigators. Safety and tolerability of pioglitazone in high-risk patients with type 2 diabetes: an overview of data from PROactive. *Drug Saf* 2009;32:187–202
 77. Schwartz AV, Chen H, Ambrosius WT, et al. Effects of TZD use and discontinuation on fracture rates in ACCORD bone study. *J Clin Endocrinol Metab* 2015;100:4059–4066
 78. Hidayat K, Du X, Wu M-J, Shi B-M. The use of metformin, insulin, sulphonylureas, and thiazolidinediones and the risk of fracture: systematic review and meta-analysis of observational studies. *Obes Rev* 2019;20:1494–1503
 79. Napoli N, Strotmeyer ES, Ensrud KE, et al. Fracture risk in diabetic elderly men: the MrOS study. *Diabetologia* 2014;57:2057–2065
 80. Hidayat K, Du X, Shi B-M. Risk of fracture with dipeptidyl peptidase-4 inhibitors, glucagon-like peptide-1 receptor agonists, or sodium-glucose cotransporter-2 inhibitors in real-world use: systematic review and meta-analysis of observational studies. *Osteoporos Int* 2019;30:1923–1940
 81. Chai S, Liu F, Yang Z, et al. Risk of fracture with dipeptidyl peptidase-4 inhibitors, glucagon-like peptide-1 receptor agonists, or sodium-glucose cotransporter-2 inhibitors in patients with type 2 diabetes mellitus: a systematic review and network meta-analysis combining 177 randomized controlled trials with a median follow-up of 26 weeks. *Front Pharmacol* 2022;13:825417
 82. Karam L, Mabileau G, Paccou J. Effects of glucagon-like peptide-1 receptor agonists on bone health in people living with obesity. *Osteoporos Int*. 8 September 2025 [Epub ahead of print]. DOI: 10.1007/s00198-025-07664-1
 83. Bilezikian JP, Watts NB, Usiskin K, et al. Evaluation of bone mineral density and bone biomarkers in patients with type 2 diabetes treated with canagliflozin. *J Clin Endocrinol Metab* 2016;101:44–51
 84. Watts NB, Bilezikian JP, Usiskin K, et al. Effects of canagliflozin on fracture risk in patients with type 2 diabetes mellitus. *J Clin Endocrinol Metab* 2016;101:157–166
 85. Perkovic V, Jardine MJ, Neal B, et al.; CREDESCENCE Trial Investigators. Canagliflozin and renal outcomes in type 2 diabetes and nephropathy. *N Engl J Med* 2019;380:2295–2306
 86. Neal B, Perkovic V, Matthews DR. Canagliflozin and cardiovascular and renal events in type 2 diabetes. *N Engl J Med* 2017;377:2099
 87. Li X, Li T, Cheng Y, et al. Effects of SGLT2 inhibitors on fractures and bone mineral density in type 2 diabetes: an updated meta-analysis. *Diabetes Metab Res Rev* 2019;35:e3170
 88. Suh S, Kim KW. Diabetes and cancer: cancer should be screened in routine diabetes assessment. *Diabetes Metab J* 2019;43:733–743
 89. Giovannucci E, Harlan DM, Archer MC, et al. Diabetes and cancer: a consensus report. *CA Cancer J Clin* 2010;60:207–221
 90. Aggarwal G, Kamada P, Chari ST. Prevalence of diabetes mellitus in pancreatic cancer compared to common cancers. *Pancreas* 2013;42:198–201
 91. Cho J, Scragg R, Petrov MS. Postpancreatitis diabetes confers higher risk for pancreatic cancer than type 2 diabetes: results from a nationwide cancer registry. *Diabetes Care* 2020;43:2106–2112
 92. Ninomiya I, Yamazaki K, Oyama K, et al. Pioglitazone inhibits the proliferation and metastasis of human pancreatic cancer cells. *Oncol Lett* 2014;8:2709–2714
 93. Hendriks AM, Schrijnders D, Kleefstra N, et al. Sulfonylurea derivatives and cancer, friend or foe? *Eur J Pharmacol* 2019;861:172598
 94. Hua Y, Zheng Y, Yao Y, Jia R, Ge S, Zhuang A. Metformin and cancer hallmarks: shedding new lights on therapeutic repurposing. *J Transl Med* 2023;21:403
 95. Wang L, Xu R, Kaelber DC, Berger NA. Glucagon-like peptide 1 receptor agonists and 13 obesity-associated cancers in patients with type 2 diabetes. *JAMA Netw Open* 2024;7:e2421305
 96. Morales DR, Bu F, Viernes B, et al. Risk of thyroid tumors with glp-1 receptor agonists: a retrospective cohort study. *Diabetes Care* 2025;48:1386–1394
 97. Xue M, Xu W, Ou Y-N, et al. Diabetes mellitus and risks of cognitive impairment and dementia: a systematic review and meta-analysis of 144 prospective studies. *Ageing Res Rev* 2019;55:100944
 98. Ohara T, Doi Y, Ninomiya T, et al. Glucose tolerance status and risk of dementia in the community: the Hisayama study. *Neurology* 2011;77:1126–1134
 99. Kciuk M, Kruczkowska W, Gałęzewska J, et al. Alzheimer's disease as type 3 diabetes: understanding the link and implications. *Int J Mol Sci* 2024;25:11955
 100. Hanyu H. Diabetes-related dementia. *Adv Exp Med Biol* 2019;1128:147–160
 101. Gudala K, Bansal D, Schifano F, Bhansali A. Diabetes mellitus and risk of dementia: a meta-analysis of prospective observational studies. *J Diabetes Investig* 2013;4:640–650
 102. Tang X, Cardoso MA, Yang J, Zhou J-B, Simó R. Impact of intensive glucose control on brain health: meta-analysis of cumulative data from 16,584 patients with type 2 diabetes mellitus. *Diabetes Ther* 2021;12:765–779
 103. Launer LJ, Miller ME, Williamson JD, et al.; ACCORD MIND investigators. Effects of intensive glucose lowering on brain structure and function in people with type 2 diabetes (ACCORD MIND): a randomised open-label substudy. *Lancet Neurol* 2011;10:969–977
 104. Xiao Y, Hong X, Neelagar R, Mo H. Association between glycated hemoglobin A1c levels, control status, and cognitive function in type 2 diabetes: a prospective cohort study. *Sci Rep* 2025;15:5011
 105. McCoy RG, Galindo RJ, Swarna KS, et al. Sociodemographic, clinical, and treatment-related factors associated with hyperglycemic crises among adults with type 1 or type 2 diabetes in the US from 2014 to 2020. *JAMA Netw Open* 2021;4:e2123471
 106. Mair ML, Athavale R, Abdelhafiz AH. Practical considerations for managing patients with diabetes

- and dementia. *Expert Rev Endocrinol Metab* 2017;12:429–440
107. Punthakee Z, Miller ME, Launer LJ, et al.; ACCORD-MIND Investigators. Poor cognitive function and risk of severe hypoglycemia in type 2 diabetes: post hoc epidemiologic analysis of the ACCORD trial. *Diabetes Care* 2012;35:787–793
108. Lacy ME, Gilsanz P, Eng C, Beeri MS, Karter AJ, Whitmer RA. Severe hypoglycemia and cognitive function in older adults with type 1 diabetes: the Study of Longevity in Diabetes (SOLID). *Diabetes Care* 2020;43:541–548
109. Lee AK, Rawlings AM, Lee CJ, et al. Severe hypoglycemia, mild cognitive impairment, dementia and brain volumes in older adults with type 2 diabetes: the Atherosclerosis Risk in Communities (ARIC) cohort study. *Diabetologia* 2018;61:1956–1965
110. Ye M, Yang Q, Zhang L, et al. Effect of hypoglycemic events on cognitive function in individuals with type 2 diabetes mellitus: a dose-response meta-analysis. *Front Neurol* 2024;15:1394499
111. Mattishent K, Loke YK. Bi-directional interaction between hypoglycaemia and cognitive impairment in elderly patients treated with glucose-lowering agents: a systematic review and meta-analysis. *Diabetes Obes Metab* 2016;18:135–141
112. Giorda CB, Ozzello A, Gentile S, et al.; HYPOS-1 Study Group of AMD. Incidence and risk factors for severe and symptomatic hypoglycemia in type 1 diabetes. Results of the HYPOS-1 study. *Acta Diabetol* 2015;52:845–853
113. Jacobson AM, Ryan CM, Braffett BH, et al.; DCCT/EDIC Research Group. Cognitive performance declines in older adults with type 1 diabetes: results from 32 years of follow-up in the DCCT and EDIC Study. *Lancet Diabetes Endocrinol* 2021;9:436–445
114. Casanova L, Hughes FJ, Preshaw PM. Diabetes and periodontal disease: a two-way relationship. *Br Dent J* 2014;217:433–437
115. Eke PI, Thornton-Evans GO, Wei L, Borgnakke WS, Dye BA, Genco RJ. Periodontitis in US adults: National Health and Nutrition Examination Survey 2009–2014. *J Am Dent Assoc* 2018;149:576–588.e6
116. Borgnakke WS, Ylöstalo PV, Taylor GW, Genco RJ. Effect of periodontal disease on diabetes: systematic review of epidemiologic observational evidence. *J Periodontol* 2013;84:S135–S152
117. Simpson TC, Clarkson JE, Worthington HV, et al. Treatment of periodontitis for glycaemic control in people with diabetes mellitus. *Cochrane Database Syst Rev* 2022;4:CD004714
118. Ziemer DC, Barnes CS, Rupasinghe RSM, Khare S. 662-P: Oral health: neglected diabetes intervention. *Diabetes* 69(Suppl 1):662-P
119. Lipman RD, O'Brien KK, Bardsley JK, Magee MF. A scoping review of the relation between toothbrushing and diabetes knowledge, glycemic control, and oral health outcomes in people with type 2 diabetes. *Diabetes Spectr* 2023;36:364–372
120. Abdul Aziz AF, Mohd-Dom TN, Mustafa N, et al. Screening for type 2 diabetes and periodontitis patients (CODAPT-MY): a multidisciplinary care approach. *BMC Health Serv Res* 2022;22:1034
121. Herrera D, Sanz M, Shapira L, et al. Association between periodontal diseases and cardiovascular diseases, diabetes and respiratory diseases: consensus report of the Joint Workshop by the European Federation of Periodontology (EFP) and the European arm of the World Organization of Family Doctors (WONCA Europe). *J Clin Periodontol* 2023;50:819–841
122. Lalla E, Papapanou PN. Diabetes mellitus and periodontitis: a tale of two common interrelated diseases. *Nat Rev Endocrinol* 2011;7:738–748
123. Christgau M, Palitzsch KD, Schmalz G, Kreiner U, Frenzel S. Healing response to non-surgical periodontal therapy in patients with diabetes mellitus: clinical, microbiological, and immunologic results. *J Clin Periodontol* 1998;25:112–124
124. Retzepi M, Donos N. The effect of diabetes mellitus on osseous healing. *Clin Oral Implants Res* 2010;21:673–681
125. United States Code. Americans with Disabilities Act of 1990. Pub. L. No. 101–336 42 U.S.C. § 2. 104 Stat. 328. p. 101–336.
126. United States Code. Americans with Disabilities Act Amendments Act of 2008. Pub. L. No. 110–325 42 U.S.C.A. § 12101 et seq
127. Wong E, Backholer K, Gearon E, et al. Diabetes and risk of physical disability in adults: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol* 2013;1:106–114
128. Tomic D, Shaw JE, Magliano DJ. The burden and risks of emerging complications of diabetes mellitus. *Nat Rev Endocrinol* 2022;18:525–539
129. Lisy K, Campbell JM, Tufanaru C, Moola S, Lockwood C. The prevalence of disability among people with cancer, cardiovascular disease, chronic respiratory disease and/or diabetes: a systematic review. *Int J Evid Based Healthc* 2018;16:154–166
130. Yang Y, Hu X, Zhang Q, Zou R. Diabetes mellitus and risk of falls in older adults: a systematic review and meta-analysis. *Age Ageing* 2016;45:761–767
131. Gregg EW, Menke A. Diabetes and Disability. In *Diabetes in America*, 3rd ed. Cowie CC, Casagrande SS, Menke A, et al., Eds. Bethesda, MD, National Institute of Diabetes and Digestive and Kidney Diseases, 2018. Available from <https://www.ncbi.nlm.nih.gov/pubmed/33651544>
132. Khan KS, Andersen H. The Impact of diabetic neuropathy on activities of daily living, postural balance and risk of falls - a systematic review. *J Diabetes Sci Technol* 2022;16:289–294
133. Elafros MA, Andersen H, Bennett DL, et al. Towards prevention of diabetic peripheral neuropathy: clinical presentation, pathogenesis, and new treatments. *Lancet Neurol* 2022;21:922–936
134. Selvarajah D, Kar D, Khunti K, et al. Diabetic peripheral neuropathy: advances in diagnosis and strategies for screening and early intervention. *Lancet Diabetes Endocrinol* 2019;7:938–948
135. Fatma S, Noohu MM. Classification of functionality of people with diabetic peripheral neuropathy based on international classification of functioning, disability and health core set (ICF-CS) of diabetes mellitus. *J Diabetes Metab Disord* 2020;19:213–221
136. Zhang Y, Lazzarini PA, McPhail SM, van Netten JJ, Armstrong DG, Pacella RE. Global disability burdens of diabetes-related lower-extremity complications in 1990 and 2016. *Diabetes Care* 2020;43:964–974
137. Tsai Y-H, Chuang L-L, Lee Y-J, Chiu C-J. How does diabetes accelerate normal aging? An examination of ADL, IADL, and mobility disability in middle-aged and older adults with and without diabetes. *Diabetes Res Clin Pract* 2021;182:109114
138. Streckmann F, Balke M, Cavaletti G, et al. Exercise and neuropathy: systematic review with meta-analysis. *Sports Med* 2022;52:1043–1065
139. Jing X, Chen J, Dong Y, et al. Related factors of quality of life of type 2 diabetes patients: a systematic review and meta-analysis. *Health Qual Life Outcomes* 2018;16:189
140. Yoon S-J, Kim K-I. Frailty and disability in diabetes. *Ann Geriatr Med Res* 2019;23:165–169
141. World Health Organization. WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). Accessed 28 August 2025. Available from <https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health/who-disability-assessment-schedule>
142. Diabetes Distress Assessment and Resource Center. Diabetes Distress Scale (DDS). Accessed 28 August 2025. Available from <https://diabetesdistress.org/take-dd-survey/>
143. Hill-Briggs F, Adler NE, Berkowitz SA, et al. Social determinants of health and diabetes: a scientific review. *Diabetes Care* 2020;44:258–279
144. Tan T-W, Shih C-D, Concha-Moore KC, et al. Correction: disparities in outcomes of patients admitted with diabetic foot infections. *PLoS One* 2019;14:e0215532
145. Skrepnek GH, Mills JL, Armstrong DG. A diabetic emergency one million feet long: disparities and burdens of illness among diabetic foot ulcer cases within emergency departments in the United States, 2006–2010. *PLoS One* 2015;10:e0134914
146. Lecube A, Hernández C, Genescà J, Simó R. Proinflammatory cytokines, insulin resistance, and insulin secretion in chronic hepatitis C patients: a case-control study. *Diabetes Care* 2006;29:1096–1101
147. Hum J, Jou JH, Green PK, et al. Improvement in glycemic control of type 2 diabetes after successful treatment of hepatitis C virus. *Diabetes Care* 2017;40:1173–1180
148. Carnovale C, Pozzi M, Dassano A, et al. The impact of a successful treatment of hepatitis C virus on glyco-metabolic control in diabetic patients: a systematic review and meta-analysis. *Acta Diabetol* 2019;56:341–354
149. Koudrat Y, Pizzol D, Cosco T, et al. High prevalence of erectile dysfunction in diabetes: a systematic review and meta-analysis of 145 studies. *Diabet Med* 2017;34:1185–1192
150. Grover SA, Lowenstein I, Kaouache M, et al. The prevalence of erectile dysfunction in the primary care setting: importance of risk factors for diabetes and vascular disease. *Arch Intern Med* 2006;166:213–219
151. Allen MS, Walter EE. Erectile dysfunction: an umbrella review of meta-analyses of risk-factors, treatment, and prevalence outcomes. *J Sex Med* 2019;16:531–541
152. Ma RC-W, So W-Y, Yang X, et al. Erectile dysfunction predicts coronary heart disease in type 2 diabetes. *J Am Coll Cardiol* 2008;51:2045–2050
153. Gazzaruso C, Solerte SB, Pujia A, et al. Erectile dysfunction as a predictor of cardiovascular events and death in diabetic patients with angiographically proven asymptomatic coronary artery disease: a potential protective role for statins and 5-phosphodiesterase inhibitors. *J Am Coll Cardiol* 2008;51:2040–2044
154. Kalter-Leibovici O, Wainstein J, Ziv A, et al.; Israel Diabetes Research Group (IDRG) Investigators.

Clinical, socioeconomic, and lifestyle parameters associated with erectile dysfunction among diabetic men. *Diabetes Care* 2005;28:1739–1744

155. De Berardis G, Pellegrini F, Franciosi M, et al.; QuED (Quality of Care and Outcomes in Type 2 Diabetes) Study Group. Longitudinal assessment of quality of life in patients with type 2 diabetes and self-reported erectile dysfunction. *Diabetes Care* 2005;28:2637–2643
156. Navaneethan SD, Vecchio M, Johnson DW, et al. Prevalence and correlates of self-reported sexual dysfunction in CKD: a meta-analysis of observational studies. *Am J Kidney Dis* 2010;56:670–685
157. Hackett G, Kirby M, Wylie K, et al. British Society for Sexual Medicine guidelines on the management of erectile dysfunction in men-2017. *J Sex Med* 2018;15:430–457
158. Corona G, Rastrelli G, Morgentaler A, Sforza A, Mannucci E, Maggi M. Meta-analysis of results of testosterone therapy on sexual function based on international index of erectile function scores. *Eur Urol* 2017;72:1000–1011
159. Dhindsa S, Miller MG, McWhirter CL, et al. Testosterone concentrations in diabetic and nondiabetic obese men. *Diabetes Care* 2010;33:1186–1192
160. Grossmann M. Low testosterone in men with type 2 diabetes: significance and treatment. *J Clin Endocrinol Metab* 2011;96:2341–2353
161. Bhasin S, Cunningham GR, Hayes FJ, et al.; Task Force, Endocrine Society. Testosterone therapy in men with androgen deficiency syndromes: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab* 2010;95:2536–2559
162. Bhasin S, Brito JP, Cunningham GR, et al. Testosterone therapy in men with hypogonadism: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab* 2018;103:1715–1744
163. Braga MAP, Rivera A, Marinheiro G, et al. Long-term cardiovascular safety of testosterone-replacement therapy in middle-aged and older men: a meta-analysis of randomized controlled trials. *Am J Cardiovasc Drugs* 2025;25:767–777
164. Shindel AW, Lue TF, Anawalt B, et al. Medical and Surgical Therapy of Erectile Dysfunction. In *Endotext*. Feingold KR, Anawalt B, Blackman MR, et al., Eds. South Dartmouth, MA, MDText.com, 2000. Available from <https://www.ncbi.nlm.nih.gov/pubmed/25905163>
165. Lindau ST, Tang H, Gomero A, et al. Sexuality among middle-aged and older adults with diagnosed and undiagnosed diabetes: a national, population-based study. *Diabetes Care* 2010;33:2202–2210
166. Shindel AW, Lue TF, Feingold KR. Sexual Dysfunction in Diabetes. In *Endotext*. Feingold KR, Anawalt B, Blackman MR, et al., Eds. South Dartmouth, MA, MDText.com, 2000. Available from <https://www.ncbi.nlm.nih.gov/pubmed/25905326>
167. Pontiroli AE, Cortelazzi D, Morabito A. Female sexual dysfunction and diabetes: a systematic review and meta-analysis. *J Sex Med* 2013;10:1044–1051
168. Van Cauwenberghe J, Enzlin P, Nefs G, et al. Prevalence of and risk factors for sexual dysfunctions in adults with type 1 or type 2 diabetes: results from Diabetes MILES - Flanders. *Diabet Med* 2022;39:e14676
169. Pyrgidis N, Mykoniatis I, Tishukov M, et al. Sexual dysfunction in women with end-stage

renal disease: a systematic review and meta-analysis. *J Sex Med* 2021;18:936–945

170. Haugstvedt A, Jørgensen J, Strandberg RB, et al. Sexual dysfunction in women with type 1 diabetes in Norway: a cross-sectional study on the prevalence and associations with physical and psychosocial complications. *Diabet Med* 2022;39:e14704
171. Buskoven MEH, Kjørholt EKH, Strandberg RB, Søfteland E, Haugstvedt A. Sexual dysfunction in women with type 1 diabetes in Norway: a qualitative study of women's experiences. *Diabet Med* 2022;39:e14856
172. Hendrieckx C, Halliday JA, Russell-Green S, et al. Adults with diabetes distress often want to talk with their health professionals about it: findings from an audit of 4 Australian specialist diabetes clinics. *Can J Diabetes* 2020;44:473–480
173. Di Stasi V, Maseroli E, Vignozzi L. Female sexual dysfunction in diabetes: mechanisms, diagnosis and treatment. *Curr Diabetes Rev* 2022;18:e171121198002
174. Wing RR, Bond DS, Gendrano IN, 3rd, et al.; Sexual Dysfunction Subgroup of the Look AHEAD Research Group. Effect of intensive lifestyle intervention on sexual dysfunction in women with type 2 diabetes: results from an ancillary Look AHEAD study. *Diabetes Care* 2013;36:2937–2944
175. Rinella ME, Lazarus JV, Ratzliff V, et al.; NAFLD Nomenclature Consensus Group. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. *Hepatology* 2023;78:1966–1986
176. Rinella ME, Neuschwander-Tetri BA, Siddiqui MS, et al. AASLD practice guidance on the clinical assessment and management of nonalcoholic fatty liver disease. *Hepatology* 2023;77:1797–1835
177. Lomonaco R, Godinez Leiva E, Bril F, et al. Advanced liver fibrosis is common in patients with type 2 diabetes followed in the outpatient setting: the need for systematic screening. *Diabetes Care* 2021;44:399–406
178. Ciardullo S, Monti T, Perseghin G. High prevalence of advanced liver fibrosis assessed by transient elastography among U.S. adults with type 2 diabetes. *Diabetes Care* 2021;44:519–525
179. Barb D, Repetto EM, Stokes ME, Shankar SS, Cusi K. Type 2 diabetes mellitus increases the risk of hepatic fibrosis in individuals with obesity and nonalcoholic fatty liver disease. *Obesity (Silver Spring)* 2021;29:1950–1960
180. Younossi ZM, Golabi P, Price JK, et al. The global epidemiology of nonalcoholic fatty liver disease and nonalcoholic steatohepatitis among patients with type 2 diabetes. *Clin Gastroenterol Hepatol* 2024;22:1999–2010
181. Stefan N, Cusi K. A global view of the interplay between non-alcoholic fatty liver disease and diabetes. *Lancet Diabetes Endocrinol* 2022;10:284–296
182. Harrison SA, Gawrieh S, Roberts K, et al. Prospective evaluation of the prevalence of non-alcoholic fatty liver disease and steatohepatitis in a large middle-aged US cohort. *J Hepatol* 2021;75:284–291
183. Castera L, Laouenan C, Vallet-Pichard A, et al.; QUID-NASH investigators. High prevalence of NASH and advanced fibrosis in type 2 diabetes: a prospective study of 330 outpatients undergoing liver biopsies for elevated ALT, using a low threshold. *Diabetes Care* 2023;46:1354–1362

184. Paik JM, Golabi P, Younossi Y, Mishra A, Younossi ZM. Changes in the global burden of chronic liver diseases from 2012 to 2017: the growing impact of NAFLD. *Hepatology* 2020;72:1605–1616
185. Simon TG, Roelstraete B, Khalili H, Hagström H, Ludvigsson JF. Mortality in biopsy-confirmed nonalcoholic fatty liver disease: results from a nationwide cohort. *Gut* 2021;70:1375–1382
186. Burra P, Becchetti C, Germani G. NAFLD and liver transplantation: disease burden, current management and future challenges. *JHEP Rep* 2020;2:100192
187. Corbin KD, Driscoll KA, Pratley RE, Smith SR, Maahs DM, Mayer-Davis EJ. Advancing Care for Type 1 Diabetes and Obesity Network (ACT1ON). Obesity in type 1 diabetes: pathophysiology, clinical impact, and mechanisms. *Endocr Rev* 2018;39:629–663
188. de Vries M, Westerink J, Kaasjager KHAH, de Valk HW. Prevalence of nonalcoholic fatty liver disease (NAFLD) in patients with type 1 diabetes mellitus: a systematic review and meta-analysis. *J Clin Endocrinol Metab* 2020;105:3842–3853
189. Cusi K, Sanyal AJ, Zhang S, et al. Non-alcoholic fatty liver disease (NAFLD) prevalence and its metabolic associations in patients with type 1 diabetes and type 2 diabetes. *Diabetes Obes Metab* 2017;19:1630–1634
190. Ciardullo S, Perseghin G. Prevalence of elevated liver stiffness in patients with type 1 and type 2 diabetes: a systematic review and meta-analysis. *Diabetes Res Clin Pract* 2022;190:109981
191. Younossi ZM, Ong JP, Takahashi H, et al.; Global Nonalcoholic Steatohepatitis Council. A global survey of physicians' knowledge about nonalcoholic fatty liver disease. *Clin Gastroenterol Hepatol* 2022;20:e1456–e1468
192. Kanwal F, Shubrook JH, Younossi Z, et al. Preparing for the NASH epidemic: a call to action. *Diabetes Care* 2021;44:2162–2172
193. Angulo P, Kleiner DE, Dam-Larsen S, et al. Liver fibrosis, but no other histologic features, is associated with long-term outcomes of patients with nonalcoholic fatty liver disease. *Gastroenterology* 2015;149:389–397
194. Ekstedt M, Hagström H, Nasr P, et al. Fibrosis stage is the strongest predictor for disease-specific mortality in NAFLD after up to 33 years of follow-up. *Hepatology* 2015;61:1547–1554
195. Taylor RS, Taylor RJ, Bayliss S, et al. Association between fibrosis stage and outcomes of patients with nonalcoholic fatty liver disease: a systematic review and meta-analysis. *Gastroenterology* 2020;158:1611–1625
196. Sanyal AJ, Van Natta ML, Clark J, et al.; NASH Clinical Research Network (CRN). Prospective study of outcomes in adults with nonalcoholic fatty liver disease. *N Engl J Med* 2021;385:1559–1569
197. Kanwal F, Shubrook JH, Adams LA, et al. Clinical care pathway for the risk stratification and management of patients with nonalcoholic fatty liver disease. *Gastroenterology* 2021;161:1657–1669
198. Gellert-Kristensen H, Richardson TG, Davey Smith G, Nordestgaard BG, Tybjaerg-Hansen A, Stender S. Combined effect of PNPLA3, TM6SF2, and HSD17B13 variants on risk of cirrhosis and hepatocellular carcinoma in the general population. *Hepatology* 2020;72:845–856

199. Stender S, Kozlitina J, Nordestgaard BG, Tybjaerg-Hansen A, Hobbs HH, Cohen JC. Adiposity amplifies the genetic risk of fatty liver disease conferred by multiple loci. *Nat Genet* 2017;49:842–847
200. Duell PB, Welty FK, Miller M, et al.; American Heart Association Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Hypertension; Council on the Kidney in Cardiovascular Disease; Council on Lifestyle and Cardiometabolic Health; and Council on Peripheral Vascular Disease. Nonalcoholic fatty liver disease and cardiovascular risk: a scientific statement from the American Heart Association. *Arterioscler Thromb Vasc Biol* 2022;42:e168–e185
201. Mantovani A, Csermely A, Petracca G, et al. Non-alcoholic fatty liver disease and risk of fatal and non-fatal cardiovascular events: an updated systematic review and meta-analysis. *Lancet Gastroenterol Hepatol* 2021;6:903–913
202. Ciardullo S, Ballabeni C, Trevisan R, Perseghin G. Liver stiffness, albuminuria and chronic kidney disease in patients with NAFLD: a systematic review and meta-analysis. *Biomolecules* 2022;12:105
203. Musso G, Gambino R, Tabibian JH, et al. Association of non-alcoholic fatty liver disease with chronic kidney disease: a systematic review and meta-analysis. *PLoS Med* 2014;11:e1001680
204. Song D, Li C, Wang Z, Zhao Y, Shen B, Zhao W. Association of non-alcoholic fatty liver disease with diabetic retinopathy in type 2 diabetic patients: a meta-analysis of observational studies. *J Diabetes Investig* 2021;12:1471–1479
205. Cusi K, Isaacs S, Barb D, et al. American Association of Clinical Endocrinology Clinical practice guideline for the diagnosis and management of nonalcoholic fatty liver disease in primary care and endocrinology clinical settings: co-sponsored by the American Association for the Study of Liver Diseases (AASLD). *Endocr Pract* 2022;28:528–562
206. Cusi K, Abdelmalek MF, Apovian CM, et al. Metabolic dysfunction-associated steatotic liver disease (MASLD) in people with diabetes: the need for screening and early intervention. A consensus report of the American Diabetes Association. *Diabetes Care* 2025;48:1057–1082
207. Eslam M, Sarin SK, Wong VW-S, et al. The Asian Pacific Association for the Study of the Liver clinical practice guidelines for the diagnosis and management of metabolic associated fatty liver disease. *Hepatol Int* 2020;14:889–919
208. European Association for the Study of the Liver (EASL); European Association for the Study of Diabetes (EASD); European Association for the Study of Obesity (EASO). EASL-EASD-EASO clinical practice guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD). *J Hepatol* 2024;81:492–542
209. Portillo-Sanchez P, Bril F, Maximos M, et al. High prevalence of nonalcoholic fatty liver disease in patients with type 2 diabetes mellitus and normal plasma aminotransferase levels. *J Clin Endocrinol Metab* 2015;100:2231–2238
210. Maximos M, Bril F, Portillo Sanchez P, et al. The role of liver fat and insulin resistance as determinants of plasma aminotransferase elevation in nonalcoholic fatty liver disease. *Hepatology* 2015;61:153–160
211. Kwo PY, Cohen SM, Lim JK. ACG clinical guideline: evaluation of abnormal liver chemistries. *Am J Gastroenterol* 2017;112:18–35
212. Younossi ZM, Anstee QM, Wai-Sun Wong V, et al. The association of histologic and noninvasive tests with adverse clinical and patient-reported outcomes in patients with advanced fibrosis due to nonalcoholic steatohepatitis. *Gastroenterology* 2021;160:1608–1619
213. Siddiqui MS, Yamada G, Vuppalanchi R, et al.; NASH Clinical Research Network. Diagnostic accuracy of noninvasive fibrosis models to detect change in fibrosis stage. *Clin Gastroenterol Hepatol* 2019;17:1877–1885
214. Unalp-Arida A, Ruhl CE. Liver fibrosis scores predict liver disease mortality in the United States population. *Hepatology* 2017;66:84–95
215. Qadri S, Ahlholm N, Lønsmann I, et al. Obesity modifies the performance of fibrosis biomarkers in nonalcoholic fatty liver disease. *J Clin Endocrinol Metab* 2022;107:e2008–e2020
216. Bril F, McPhaul MJ, Caulfield MP, et al. Performance of plasma biomarkers and diagnostic panels for nonalcoholic steatohepatitis and advanced fibrosis in patients with type 2 diabetes. *Diabetes Care* 2020;43:290–297
217. McPherson S, Hardy T, Dufour J-F, et al. Age as a confounding factor for the accurate non-invasive diagnosis of advanced NAFLD fibrosis. *Am J Gastroenterol* 2017;112:740–751
218. Castera L, Friedrich-Rust M, Loomba R. Noninvasive assessment of liver disease in patients with nonalcoholic fatty liver disease. *Gastroenterology* 2019;156:1264–1281
219. Eddowes PJ, Sasso M, Allison M, et al. Accuracy of FibroScan controlled attenuation parameter and liver stiffness measurement in assessing steatosis and fibrosis in patients with nonalcoholic fatty liver disease. *Gastroenterology* 2019;156:1717–1730
220. Mózes FE, Lee JA, Selvaraj EA, et al.; LITMUS Investigators. Diagnostic accuracy of non-invasive tests for advanced fibrosis in patients with NAFLD: an individual patient data meta-analysis. *Gut* 2022;71:1006–1019
221. Elhence A, Anand A, Biswas S, et al. Compensated advanced chronic liver disease in nonalcoholic fatty liver disease: two-step strategy is better than Baveno criteria. *Dig Dis Sci* 2023;68:1016–1025
222. Wong VW, Zelber-Sagi S, Cusi K, et al. Management of NAFLD in primary care settings. *Liver Int* 2022;42:2377–2389
223. Lazarus JV, Anstee QM, Hagström H, et al. Defining comprehensive models of care for NAFLD. *Nat Rev Gastroenterol Hepatol* 2021;18:717–729
224. Cusi K, Budd J, Johnson E, Shubrook J. Making sense of the nonalcoholic fatty liver disease clinical practice guidelines: what clinicians need to know. *Diabetes Spectr* 2024;37:29–38
225. Anstee QM, Lawitz EJ, Alkhouri N, et al. Noninvasive tests accurately identify advanced fibrosis due to NASH: baseline data from the STELLAR trials. *Hepatology* 2019;70:1521–1530
226. Lee J, Vali Y, Boursier J, et al. Prognostic accuracy of FIB-4, NAFLD fibrosis score and APRI for NAFLD-related events: a systematic review. *Liver Int* 2021;41:261–270
227. Chan W-K, Treeprasertsuk S, Goh GB-B, et al. Optimizing use of nonalcoholic fatty liver disease fibrosis score, fibrosis-4 score, and liver stiffness measurement to identify patients with advanced fibrosis. *Clin Gastroenterol Hepatol* 2019;17:2570–2580
228. Petta S, Wai-Sun Wong V, Bugianesi E, et al. Impact of obesity and alanine aminotransferase levels on the diagnostic accuracy for advanced liver fibrosis of noninvasive tools in patients with nonalcoholic fatty liver disease. *Am J Gastroenterol* 2019;114:916–928
229. Vali Y, Lee J, Boursier J, et al.; LITMUS systematic review team. Enhanced liver fibrosis test for the non-invasive diagnosis of fibrosis in patients with NAFLD: a systematic review and meta-analysis. *J Hepatol* 2020;73:252–262
230. Srivastava A, Gailer R, Tanwar S, et al. Prospective evaluation of a primary care referral pathway for patients with non-alcoholic fatty liver disease. *J Hepatol* 2019;71:371–378
231. Saarinen K, Färkkilä M, Jula A, et al. Enhanced liver fibrosis test predicts liver-related outcomes in the general population. *JHEP Rep* 2023;5:100765
232. Kjaergaard M, Lindvig KP, Thorhauge KH, et al. Using the ELF test, FIB-4 and NAFLD fibrosis score to screen the population for liver disease. *J Hepatol* 2023;79:277–286
233. Zoncapè M, Liguori A, Tsochatzis EA. Non-invasive testing and risk-stratification in patients with MASLD. *Eur J Intern Med* 2024;122:11–19
234. Andersson A, Kelly M, Imajo K, et al. Clinical utility of magnetic resonance imaging biomarkers for identifying nonalcoholic steatohepatitis patients at high risk of progression: a multicenter pooled data and meta-analysis. *Clin Gastroenterol Hepatol* 2022;20:2451–2461
235. Long MT, Noureddin M, Lim JK. AGA clinical practice update: diagnosis and management of nonalcoholic fatty liver disease in lean individuals: expert review. *Gastroenterology* 2022;163:764–774
236. Cusi K. Role of obesity and lipotoxicity in the development of nonalcoholic steatohepatitis: pathophysiology and clinical implications. *Gastroenterology* 2012;142:711–725
237. Loomba R, Friedman SL, Shulman GI. Mechanisms and disease consequences of nonalcoholic fatty liver disease. *Cell* 2021;184:2537–2564
238. Schuppan D, Surabattula R, Wang XY. Determinants of fibrosis progression and regression in NASH. *J Hepatol* 2018;68:238–250
239. Promrat K, Kleiner DE, Niemeier HM, et al. Randomized controlled trial testing the effects of weight loss on nonalcoholic steatohepatitis. *Hepatology* 2010;51:121–129
240. Vilar-Gomez E, Martinez-Perez Y, Calzadilla-Bertot L, et al. Weight loss through lifestyle modification significantly reduces features of nonalcoholic steatohepatitis. *Gastroenterology* 2015;149:367–378
241. Younossi ZM, Zelber-Sagi S, Henry L, Gerber LH. Lifestyle interventions in nonalcoholic fatty liver disease. *Nat Rev Gastroenterol Hepatol* 2023;20:708–722
242. Arab JP, Dirchwolf M, Álvares-da-Silva MR, et al. Latin American Association for the Study of the Liver (ALEH) practice guidance for the diagnosis and treatment of non-alcoholic fatty liver disease. *Ann Hepatol* 2020;19:674–690
243. Sargeant JA, Gray LJ, Bodicoat DH, et al. The effect of exercise training on intrahepatic triglyceride and hepatic insulin sensitivity: a systematic review and meta-analysis. *Obes Rev* 2018;19:1446–1459
244. Kanwal F, Kramer JR, Li L, et al. Effect of metabolic traits on the risk of cirrhosis and

- hepatocellular cancer in nonalcoholic fatty liver disease. *Hepatology* 2020;71:808–819
245. Younossi Z, Stepanova M, Sanyal AJ, et al. The conundrum of cryptogenic cirrhosis: adverse outcomes without treatment options. *J Hepatol* 2018;69:1365–1370
246. Castera L, Cusi K. Diabetes and cirrhosis: current concepts on diagnosis and management. *Hepatology* 2023;77:2128–2146
247. Patel Chavez C, Cusi K, Kadiyala S. The emerging role of glucagon-like peptide-1 receptor agonists for the management of NAFLD. *J Clin Endocrinol Metab* 2022;107:29–38
248. Gastaldelli A, Cusi K. From NASH to diabetes and from diabetes to NASH: mechanisms and treatment options. *JHEP Rep* 2019;1:312–328
249. Budd J, Cusi K. Role of agents for the treatment of diabetes in the management of nonalcoholic fatty liver disease. *Curr Diab Rep* 2020;20:59
250. Gonzalez-Rellan MJ, Drucker DJ. The expanding benefits of GLP-1 medicines. *Cell Rep Med* 2025;6:102214
251. Musso G, Cassader M, Paschetta E, Gambino R. Thiazolidinediones and Advanced Liver Fibrosis in Nonalcoholic Steatohepatitis: a Meta-analysis. *JAMA Intern Med* 2017;177:633–640
252. Bril F, Kalavalapalli S, Clark VC, et al. Response to pioglitazone in patients with nonalcoholic steatohepatitis with vs without type 2 diabetes. *Clin Gastroenterol Hepatol* 2018;16:558–566
253. Sanyal AJ, Newsome PN, Kliers I, et al.; ESSENCE Study Group. Phase 3 trial of semaglutide in metabolic dysfunction-associated steatohepatitis. *N Engl J Med* 2025;392:2089–2099
254. U.S. Food and Drug Administration. FDA Approves Treatment for Serious Liver Disease Known as ‘MASH’. Accessed 28 August 2025. Available from <https://www.fda.gov/drugs/news-events-human-drugs/fda-approves-treatment-serious-liver-disease-known-mash>
255. Belfort R, Harrison SA, Brown K, et al. A placebo-controlled trial of pioglitazone in subjects with nonalcoholic steatohepatitis. *N Engl J Med* 2006;355:2297–2307
256. Huang J-F, Dai C-Y, Huang C-F, et al. First-in-Asian double-blind randomized trial to assess the efficacy and safety of insulin sensitizer in nonalcoholic steatohepatitis patients. *Hepatol Int* 2021;15:1136–1147
257. Cusi K, Orsak B, Bril F, et al. Long-term pioglitazone treatment for patients with nonalcoholic steatohepatitis and prediabetes or type 2 diabetes mellitus: a randomized trial. *Ann Intern Med* 2016;165:305–315
258. Sanyal AJ, Chalasani N, Kowdley KV, et al.; NASH CRN. Pioglitazone, vitamin E, or placebo for nonalcoholic steatohepatitis. *N Engl J Med* 2010;362:1675–1685
259. Aithal GP, Thomas JA, Kaye PV, et al. Randomized, placebo-controlled trial of pioglitazone in nondiabetic subjects with nonalcoholic steatohepatitis. *Gastroenterology* 2008;135:1176–1184
260. Sathyanarayana P, Jogi M, Muthupillai R, Krishnamurthy R, Samson SL, Bajaj M. Effects of combined exenatide and pioglitazone therapy on hepatic fat content in type 2 diabetes. *Obesity (Silver Spring)* 2011;19:2310–2315
261. Abdul-Ghani M, Megahed A, DeFronzo RA, Al-Ozairi E, Jayyousi A. Combination therapy with pioglitazone/exenatide improves beta-cell function and produces superior glycaemic control compared with basal/bolus insulin in poorly controlled type 2 diabetes: a 3-year follow-up of the Qatar study. *Diabetes Obes Metab* 2020;22:2287–2294
262. Lavynenko O, Abdul-Ghani M, Alatrach M, et al. Combination therapy with pioglitazone/exenatide/metformin reduces the prevalence of hepatic fibrosis and steatosis: the efficacy and durability of initial combination therapy for type 2 diabetes (EDICT). *Diabetes Obes Metab* 2022;24:899–907
263. Ahrén B, Masmiquel L, Kumar H, et al. Efficacy and safety of once-weekly semaglutide versus once-daily sitagliptin as an add-on to metformin, thiazolidinediones, or both, in patients with type 2 diabetes (SUSTAIN 2): a 56-week, double-blind, phase 3a, randomised trial. *Lancet Diabetes Endocrinol* 2017;5:341–354
264. Noureddin M, Jones C, Alkhoury N, Gomez EV, Dieterich DT, Rinella ME; NASHNET. Screening for nonalcoholic fatty liver disease in persons with type 2 diabetes in the United States is cost-effective: a comprehensive cost-utility analysis. *Gastroenterology* 2020;159:1985–1987
265. Mahady SE, Wong G, Craig JC, George J. Pioglitazone and vitamin E for nonalcoholic steatohepatitis: a cost utility analysis. *Hepatology* 2012;56:2172–2179
266. Kovacs CS, Seshiah V, Swallow R, et al.; EMPA-REG PIO trial investigators. Empagliflozin improves glycaemic and weight control as add-on therapy to pioglitazone or pioglitazone plus metformin in patients with type 2 diabetes: a 24-week, randomized, placebo-controlled trial. *Diabetes Obes Metab* 2014;16:147–158
267. Armstrong MJ, Gaunt P, Aithal GP, et al.; LEAN trial team. Liraglutide safety and efficacy in patients with non-alcoholic steatohepatitis (LEAN): a multicentre, double-blind, randomised, placebo-controlled phase 2 study. *Lancet* 2016;387:679–690
268. Gastaldelli A, Cusi K, Fernández Landó L, Bray R, Brouwers B, Rodríguez Á. Effect of tirzepatide versus insulin degludec on liver fat content and abdominal adipose tissue in people with type 2 diabetes (SURPASS-3 MRI): a substudy of the randomised, open-label, parallel-group, phase 3 SURPASS-3 trial. *Lancet Diabetes Endocrinol* 2022;10:393–406
269. Loomba R, Hartman ML, Lawitz EJ, et al.; SYNERGY-NASH Investigators. Tirzepatide for metabolic dysfunction-associated steatohepatitis with liver fibrosis. *N Engl J Med* 2024;391:299–310
270. Sanyal AJ, Bedossa P, Fraessdorf M, et al.; 1404-0043 Trial Investigators. A phase 2 randomized trial of survodutide in MASH and fibrosis. *N Engl J Med* 2024;391:311–319
271. Cusi K, Bril F, Barb D, et al. Effect of canagliflozin treatment on hepatic triglyceride content and glucose metabolism in patients with type 2 diabetes. *Diabetes Obes Metab* 2019;21:812–821
272. Kahl S, Gancheva S, Straßburger K, et al. Empagliflozin effectively lowers liver fat content in well-controlled type 2 diabetes: a randomized, double-blind, phase 4, placebo-controlled trial. *Diabetes Care* 2020;43:298–305
273. Latva-Rasku A, Honka M-J, Kullberg J, et al. The SGLT2 inhibitor dapagliflozin reduces liver fat but does not affect tissue insulin sensitivity: a randomized, double-blind, placebo-controlled study with 8-week treatment in type 2 diabetes patients. *Diabetes Care* 2019;42:931–937
274. Harrison SA, Bedossa P, Guy CD, et al.; MAESTRO-NASH Investigators. A phase 3, randomized, controlled trial of resmetirom in NASH with liver fibrosis. *N Engl J Med* 2024;390:497–509
275. Cusi K. Selective agonists of thyroid hormone receptor beta for the treatment of NASH. *N Engl J Med* 2024;390:559–561
276. Chen VL, Morgan TR, Rotman Y, et al. Resmetirom therapy for metabolic dysfunction-associated steatotic liver disease: October 2024 updates to AASLD Practice Guidance [published correction appears in *Hepatology* 2025;81:E133]. *Hepatology* 2025;81:312–320
277. Loomba R, Abdelmalek MF, Armstrong MJ, et al.; NN9931-4492 investigators. Semaglutide 2.4 mg once weekly in patients with non-alcoholic steatohepatitis-related cirrhosis: a randomised, placebo-controlled phase 2 trial. *Lancet Gastroenterol Hepatol* 2023;8:511–522
278. Aminian A, Al-Kurd A, Wilson R, et al. Association of bariatric surgery with major adverse liver and cardiovascular outcomes in patients with biopsy-proven nonalcoholic steatohepatitis. *JAMA* 2021;326:2031–2042
279. Fakhry TK, Mhaskar R, Schwitalla T, Muradova E, Gonzalvo JP, Murr MM. Bariatric surgery improves nonalcoholic fatty liver disease: a contemporary systematic review and meta-analysis. *Surg Obes Relat Dis* 2019;15:502–511
280. Ramai D, Singh J, Lester J, et al. Systematic review with meta-analysis: bariatric surgery reduces the incidence of hepatocellular carcinoma. *Aliment Pharmacol Ther* 2021;53:977–984
281. Choi J, Nguyen VH, Przybyszewski E, et al. Statin use and risk of hepatocellular carcinoma and liver fibrosis in chronic liver disease. *JAMA Intern Med* 2025;185:522–530
282. Kaplan DE, Serper MA, Mehta R, et al.; VOCAL Study Group. Effects of hypercholesterolemia and statin exposure on survival in a large national cohort of patients with cirrhosis. *Gastroenterology* 2019;156:1693–1706
283. Li C, Ford ES, Zhao G, Croft JB, Balluz LS, Mokdad AH. Prevalence of self-reported clinically diagnosed sleep apnea according to obesity status in men and women: national Health and Nutrition Examination Survey, 2005–2006. *Prev Med* 2010;51:18–23
284. West SD, Nicoll DJ, Stradling JR. Prevalence of obstructive sleep apnoea in men with type 2 diabetes. *Thorax* 2006;61:945–950
285. Resnick HE, Redline S, Shahar E, et al.; Sleep Heart Health Study. Diabetes and sleep disturbances: findings from the Sleep Heart Health Study. *Diabetes Care* 2003;26:702–709
286. Foster GD, Sanders MH, Millman R, et al.; Sleep AHEAD Research Group. Obstructive sleep apnea among obese patients with type 2 diabetes. *Diabetes Care* 2009;32:1017–1019
287. Bibbins-Domingo K, Grossman DC, Curry SJ, et al.; US Preventive Services Task Force. Screening for obstructive sleep apnea in adults: US Preventive Services Task Force recommendation statement. *JAMA* 2017;317:407–414
288. Malhotra A, Grunstein RR, Fietze I, et al.; SURMOUNT-OSA Investigators. Tirzepatide for the treatment of obstructive sleep apnea and obesity. *N Engl J Med* 2024;391:1193–1205
289. Piciucchi M, Capurso G, Archibugi L, Delle Fave MM, Capasso M, Delle Fave G. Exocrine pancreatic insufficiency in diabetic patients:

- prevalence, mechanisms, and treatment. *Int J Endocrinol* 2015;2015:595649
290. Lee Y-K, Huang M-Y, Hsu C-Y, Su Y-C. Bidirectional relationship between diabetes and acute pancreatitis: a population-based cohort study in Taiwan. *Medicine (Baltimore)* 2016;95:e2448
291. Das SLM, Singh PP, Phillips ARJ, Murphy R, Windsor JA, Petrov MS. Newly diagnosed diabetes mellitus after acute pancreatitis: a systematic review and meta-analysis. *Gut* 2014;63:818–831
292. Petrov MS. Diabetes of the exocrine pancreas: American Diabetes Association-compliant lexicon. *Pancreatol* 2017;17:523–526
293. Thomsen RW, Pedersen L, Møller N, Kahlert J, Beck-Nielsen H, Sørensen HT. Incretin-based therapy and risk of acute pancreatitis: a nationwide population-based case-control study. *Diabetes Care* 2015;38:1089–1098
294. Tkáč I, Raz I. Combined analysis of three large interventional trials with gliptins indicates increased incidence of acute pancreatitis in patients with type 2 diabetes. *Diabetes Care* 2017;40:284–286
295. Sutherland DER, Radosevich DM, Bellin MD, et al. Total pancreatectomy and islet autotransplantation for chronic pancreatitis. *J Am Coll Surg* 2012;214:409–424
296. Quartuccio M, Hall E, Singh V, et al. Glycemic predictors of insulin independence after total pancreatectomy with islet autotransplantation. *J Clin Endocrinol Metab* 2017;102:801–809
297. Webb MA, Illouz SC, Pollard CA, et al. Islet auto transplantation following total pancreatectomy: a long-term assessment of graft function. *Pancreas* 2008;37:282–287
298. Wu Q, Zhang M, Qin Y, et al. Systematic review and meta-analysis of islet autotransplantation after total pancreatectomy in chronic pancreatitis patients. *Endocr J* 2015;62:227–234
299. Baiduc RR, Helzner EP. Epidemiology of diabetes and hearing loss. *Semin Hear* 2019;40:281–291
300. Helzner EP, Contrera KJ. Type 2 diabetes and hearing impairment. *Curr Diab Rep* 2016;16:3
301. Hicks CW, Wang D, Lin FR, Reed N, Windham BG, Selvin E. Peripheral neuropathy and vision and hearing impairment in US adults with and without diabetes. *Am J Epidemiol* 2023;192:237–245
302. Bainbridge KE, Hoffman HJ, Cowie CC. Risk factors for hearing impairment among U.S. adults with diabetes: National Health and Nutrition Examination Survey 1999–2004. *Diabetes Care* 2011;34:1540–1545
303. Schade DS, Lorenzi GM, Braffett BH, et al.; DCCT/EDIC Research Group. Hearing impairment and type 1 diabetes in the Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complications (DCCT/EDIC) cohort. *Diabetes Care* 2018;41:2495–2501
304. Rasmussen VF, Vestergaard ET, Hejlesen O, Andersson CUN, Cichosz SL. Prevalence of taste and smell impairment in adults with diabetes: a cross-sectional analysis of data from the National Health and Nutrition Examination Survey (NHANES). *Prim Care Diabetes* 2018;12:453–459
305. Centers for Disease Control and Prevention. Interim Clinical Considerations for Use of COVID-19 Vaccines in the United States. Accessed 19 September 2025. Available from <https://www.cdc.gov/vaccines/covid-19/clinical-considerations/covid-19-vaccines-us.html>
306. Sandul AL, Rapposelli K, Nyendak M, Kim M. Updated recommendation for universal hepatitis b vaccination in adults aged 19–59 years - United States, 2024. *MMWR Morb Mortal Wkly Rep* 2024;73:1106
307. Davies MJ, Aroda VR, Collins BS, et al. Management of hyperglycemia in type 2 diabetes, 2022. A consensus report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). *Diabetes Care* 2022;45:2753–2786